

Rubin 2036

Lightning talks

Deep-Learning Based Detection, Instance Segmentation & Star-Galaxy Classification with DeepDISC for LSST DP1

By Prince Yadav

Advisor: Prof. Xin Liu



1. Input

Real
DP1 Tile

2. Backbone

Swin-B
↓

3. FPN

↑

4. RPN

Top-Down Region
Proposals

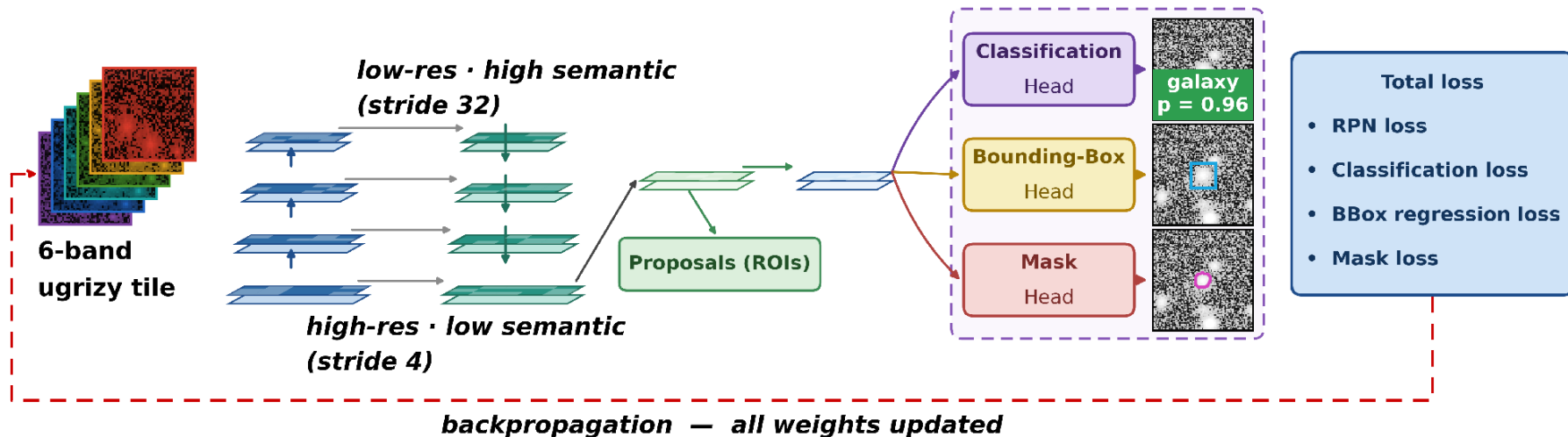
5. RoIAlign

Fixed-Size
Feats

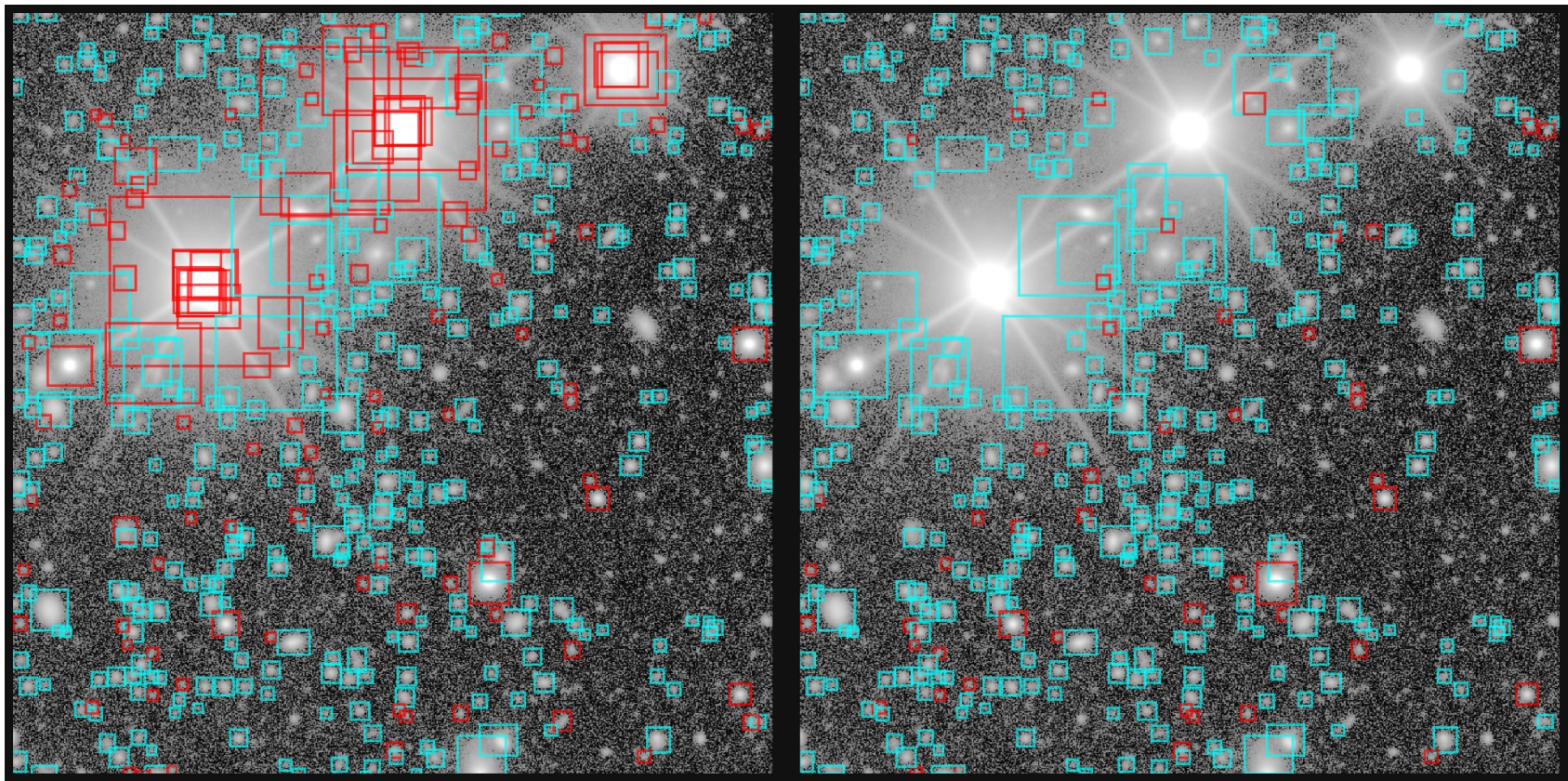
6. ROI cascade heads

3-Stage Cascade
IoU 0.5-0.7

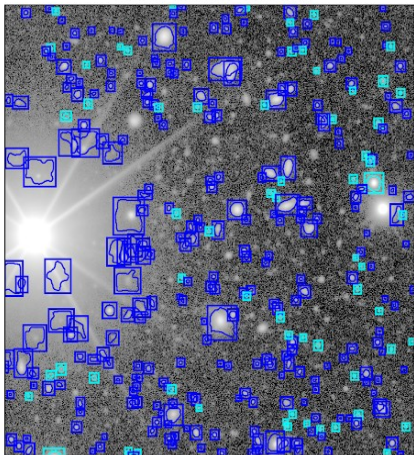
7. Loss



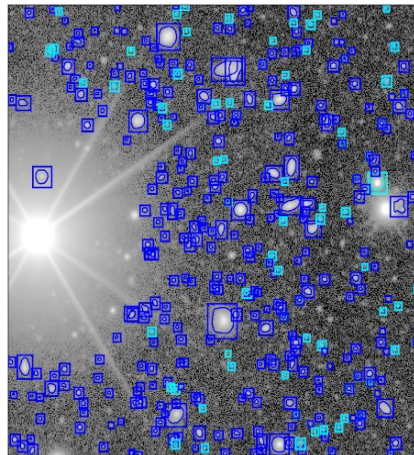
Train: Scarlet curated ground truth + refExtendedness



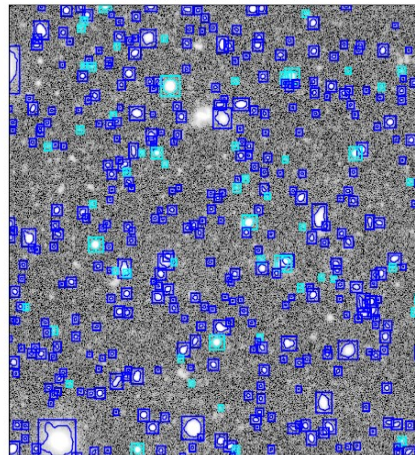
**Scarlet curated pseudo
ground truth**



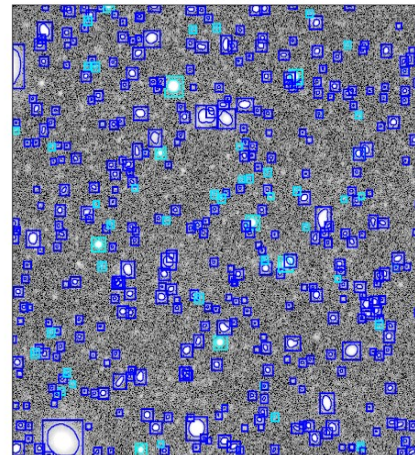
Deepdisc Predictions



**Scarlet curated pseudo
ground truth**



**Deepdisc
Predictions**



Some

Most



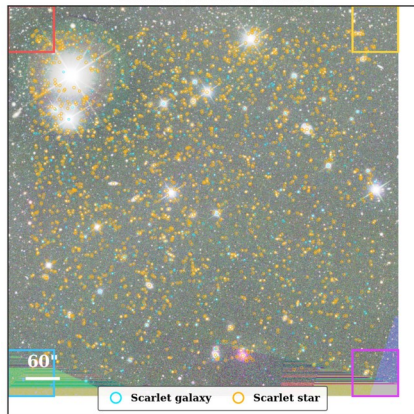
galaxy



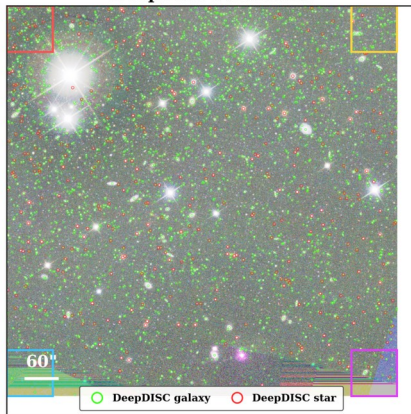
star

Tract 5063 · Patch 46 · ECDFS
RA $\approx 52.805^\circ$, Dec $\approx -27.604^\circ$ ($\sim 11.3'$ across)

Scarlet Ground Truth



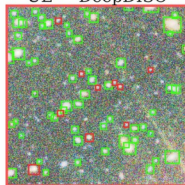
DeepDISC Prediction



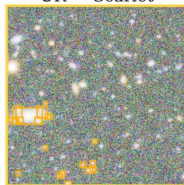
UL · Scarlet



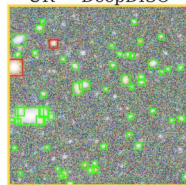
UL · DeepDISC



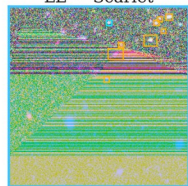
UR · Scarlet



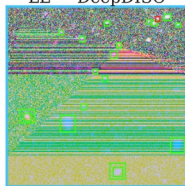
UR · DeepDISC



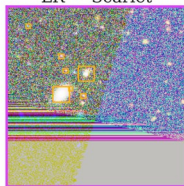
LL · Scarlet



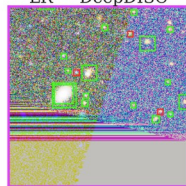
LL · DeepDISC



LR · Scarlet

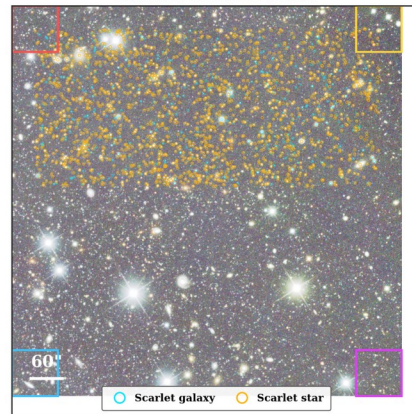


LR · DeepDISC

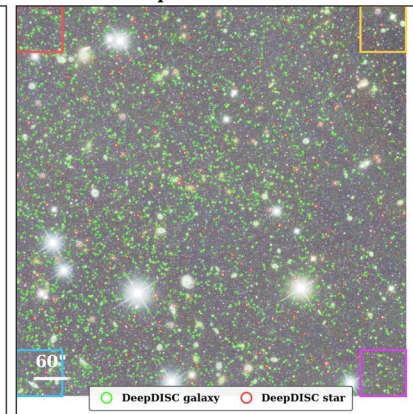


Tract 4848 · Patch 92 · ECDFS
RA $\approx 52.622^\circ$, Dec $\approx -28.257^\circ$ ($\sim 11.3'$ across)

Scarlet Ground Truth



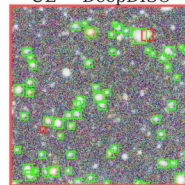
DeepDISC Prediction



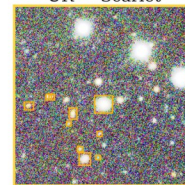
UL · Scarlet



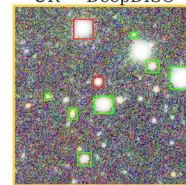
UL · DeepDISC



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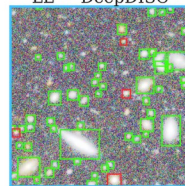
UR · DeepDISC



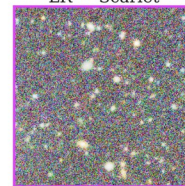
LL · Scarlet



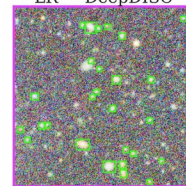
LL · DeepDISC



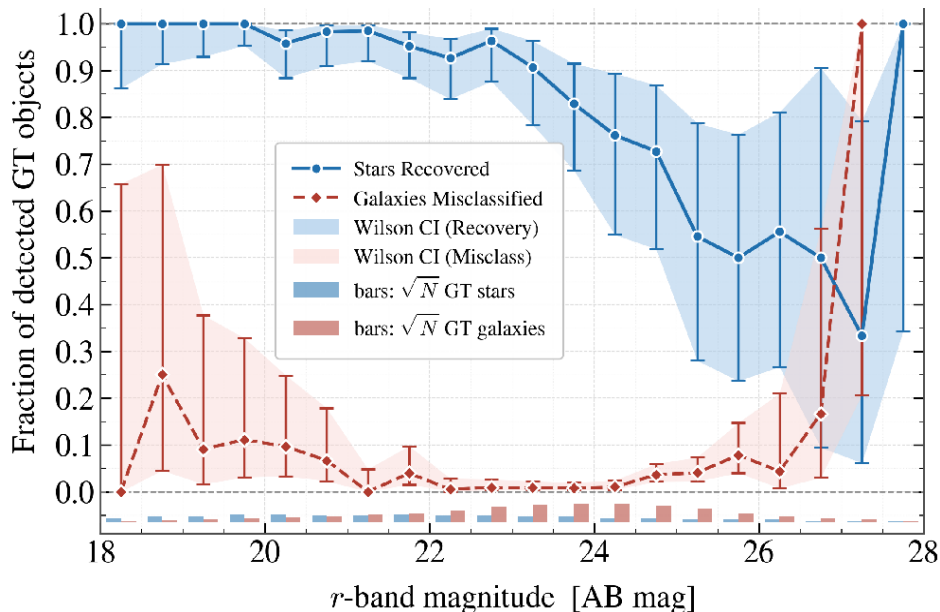
LR · Scarlet



LR · DeepDISC



Configuration	Galaxy	Star
DP1 FT (headline)	84.9	74.3
+ star-weighted	84.6	76.6
+ star-weighted + focal	87.0	78.2



***FOR Classification,
DeepDISC reaches Near-parity with RF
& Far exceeds morphology-only classification**

External Validation — HST / CANDELS

Metric %	DeepDISC*	DP1/RSP
Completeness	31.0	43.8
Purity	94.6	75.4
HST-anchored UB	14.9	18.0
Galaxy recall	94.8	85.3
Star recall	71.4	61.5

**leakage-free held-out footprint (94.6% purity)
and lower UB than the production catalog.*

[DESC-PUB-00258]

DeepDISC on LSST Data Preview 1:

Scene-Level Detection, Deblending,
and Star/Galaxy Classification
in the ECDFS and EDFS Fields

Astro2020 APC White Paper

Next Generation LSST Science

<https://arxiv.org/abs/1907.08945>

Principal Author:

Name: Saurabh W. Jha

Institution: Rutgers, the State University of New Jersey

Email: saurabh@physics.rutgers.edu

Phone: +1 848-445-8962

Co-authors:

Federica Bianco, University of Delaware

W. Niel Brandt, Pennsylvania State University

Gaspar Galaz, Pontificia Universidad Católica de Chile

Eric Gawiser, Rutgers University

John Gizis, University of Delaware

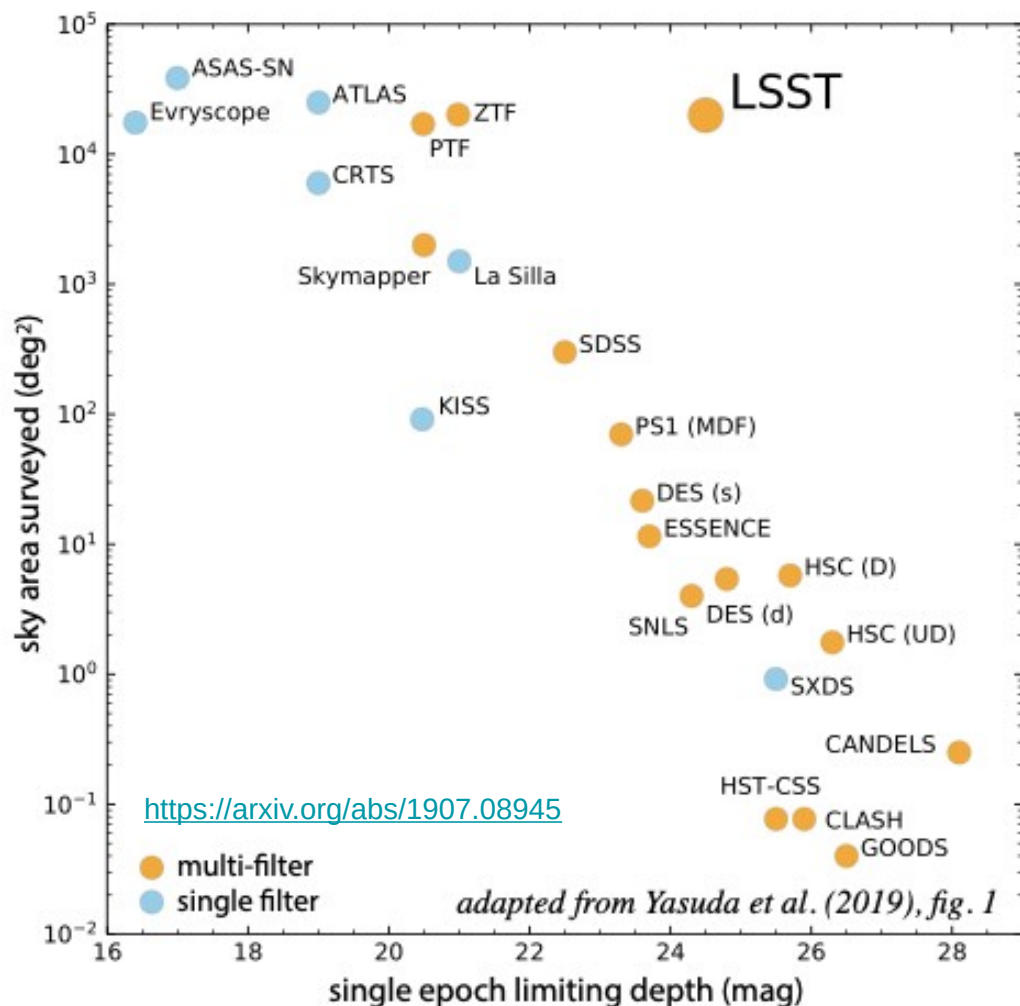
Renée Hložek, University of Toronto

Sugata Kaviraj, University of Hertfordshire

Jeffrey A. Newman, University of Pittsburgh

Aprajita Verma, University of Oxford

W. Michael Wood-Vasey, University of Pittsburgh



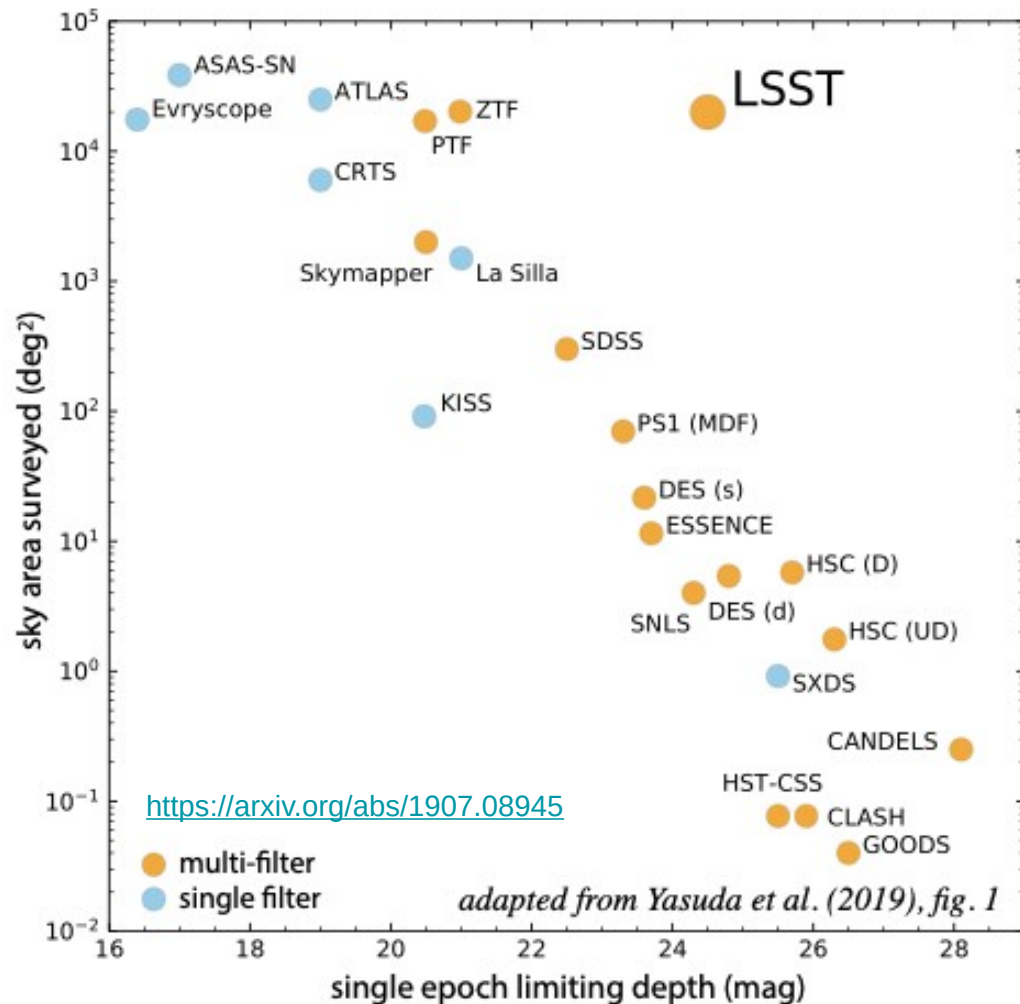
LSST is revolutionary for time-domain

wide range of frontier science: solar system, exoplanets, stellar variables and microlensing, standard candles, novae, supernovae, kilonovae/GW counterparts, GRBs, AGN, lens time delays, ...

Celestial cinematography with Rubin

Rubin is the chronicler of the night sky
the movie of the Universe
does not end in 2036!

we will need a Rubin-like observatory for transient discovery... so why not Rubin?



Leveraging Rubin for the time domain

large aperture, wide-field, and *fast*
so **not** the facility for sit & stare long exposures,
e.g., multi-object spectroscopy

narrow-band or medium-band filters will reduce the
transient discovery rate and aren't ideal for light curves

shifted broad-band filters as the workhorse:

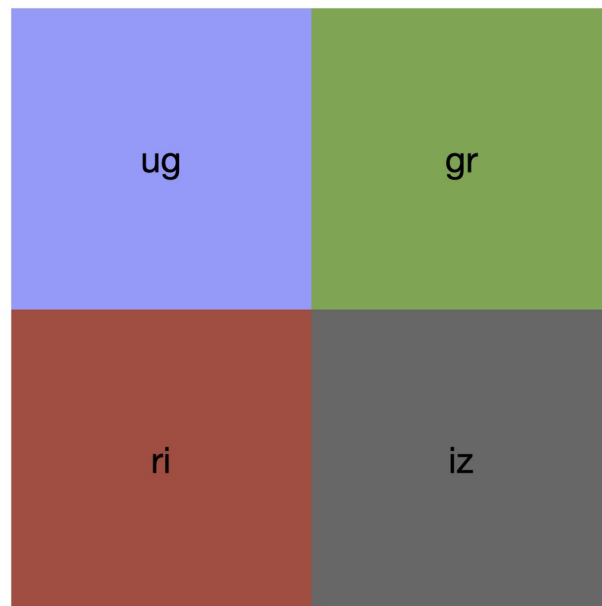
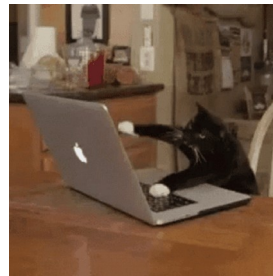
would keep a high transient rate, good light curves, and
benefit static science (e.g., photo-z; Jeff Newman talk!)

Rubin + LSSTCam: *fast slew*, but *slow filter change*:

patterned filter to tile sky

efficient (e.g., TOOs), less wear & tear, frees up slots

a look into the control room...
imaging spectroscopy



some more general thoughts

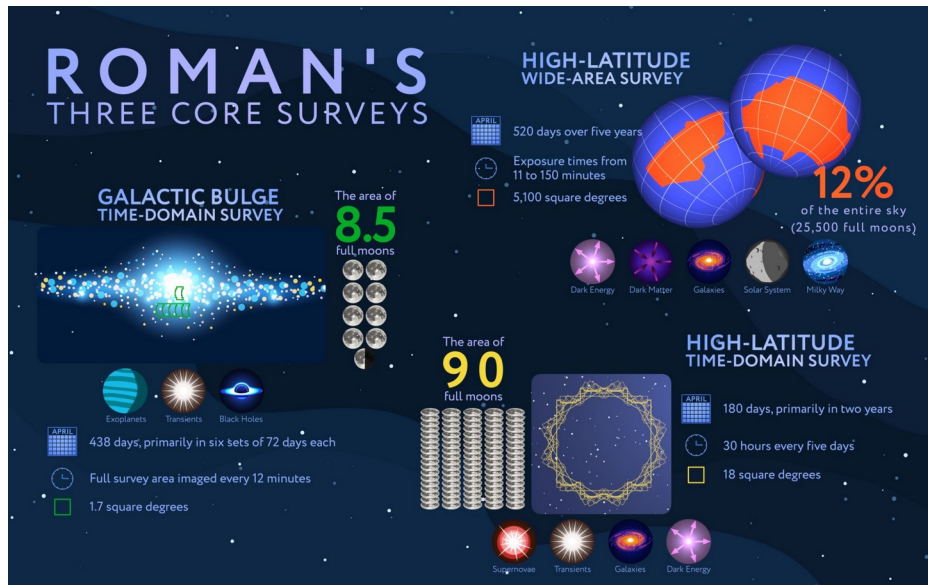
experience from SCOC (even just DDFs!),
cadence notes, white papers, etc.
many more survey ideas than survey time

what will key science questions be in 2036?

community-driven science
community-defined surveys

include proposal-based surveys of all scales

Roman process can be a model



Revisiting Quadratic Scalar Field (Fuzzy, Wave, Ultralight) Dark Matter in Light of DESI DR2 and Current Cosmological Observations

Muhammad Awais

Escuela Superior de Física y Matemáticas ESFM | IPN Oficial

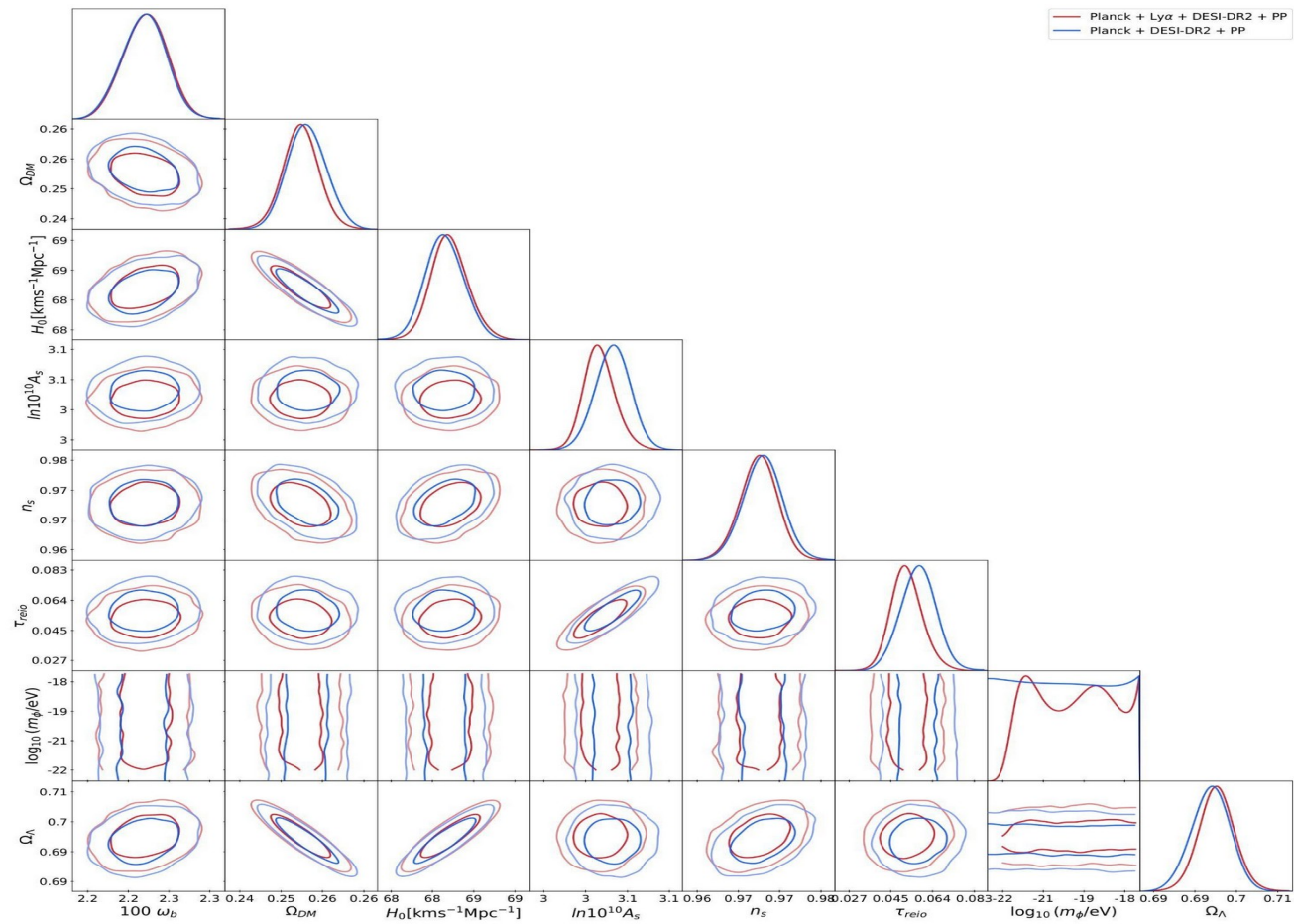
Λ CDM successfully describes the large-scale Universe using CMB, BAO, and SN Ia data.

- ▶ However, the physical nature of dark matter remains unknown.
- ▶ Ultra-light scalar-field dark matter is a motivated alternative that can modify structure formation on small scales .
- ▶ In the quadratic model,
 $V(\phi) = \frac{1}{2} m_\phi^2 \phi^2$,
the field behaves like pressureless matter after coherent oscillations begin.

Main question

Can current cosmological observations place meaningful constraints on the scalar-field mass m_ϕ ?

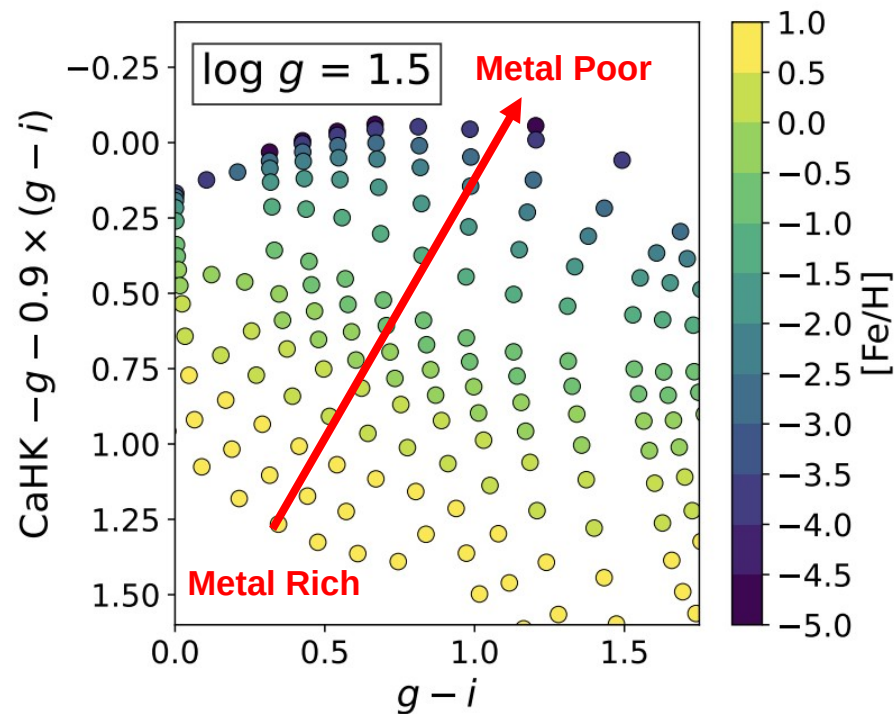
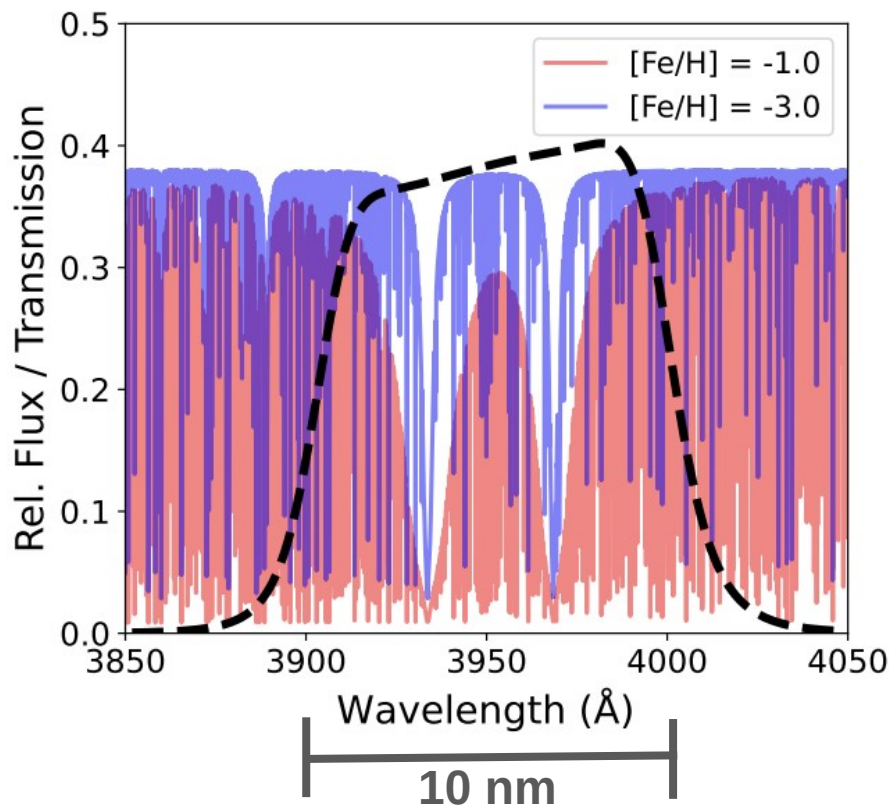
Results



Quadratic SFDM remains compatible with current cosmological observations.

- ▶ Its main signature is the suppression of the linear matter power spectrum on small scales.
- ▶ The CMB temperature spectrum remains very close to Λ CDM.
- ▶ Baseline data alone do not strongly constrain the scalar-field mass.
- ▶ In this analysis, Ly α information gives the strongest model-specific constraint: **$\log_{10}(m_{\phi}/\text{eV}) > -22.15$ (95% C.L.)**

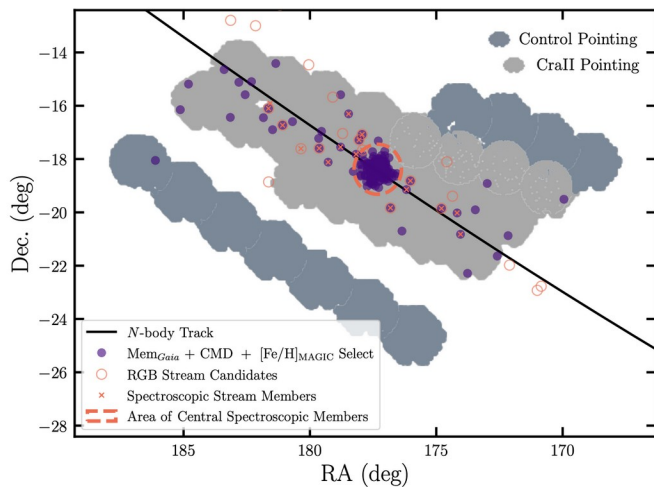
Metallicity-Sensitive Imaging for Dark Matter Science



DECam MAGIC Survey
Chiti et al. [2605.26581]

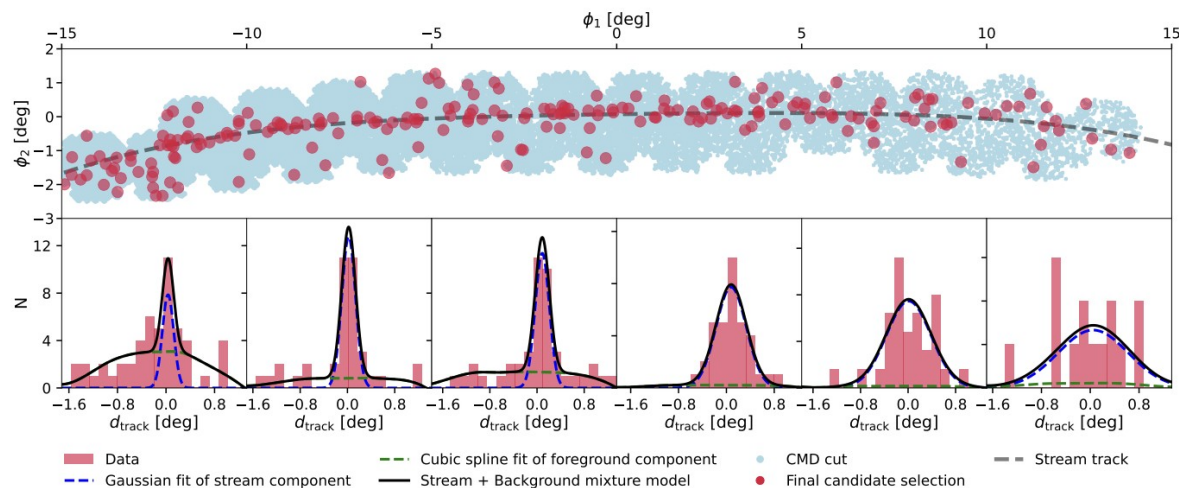
Metallicity-Sensitive Imaging for Dark Matter Science

Discovery of tidal tails around
the disrupting dwarf galaxy Crater II



Atzberger, Pace et al. [2604.13374]

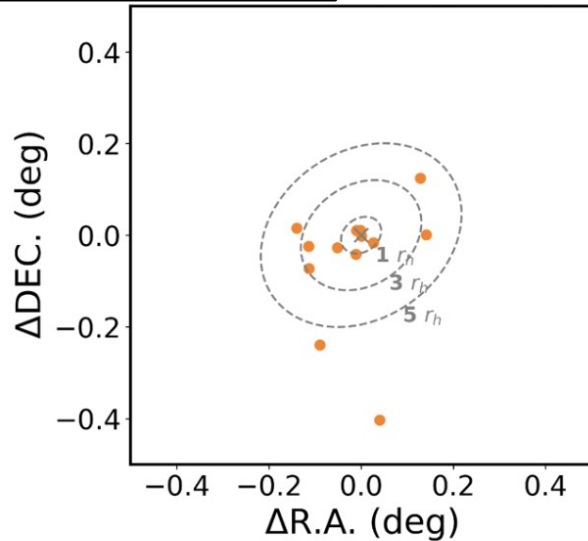
Cocoon around the Jet stream



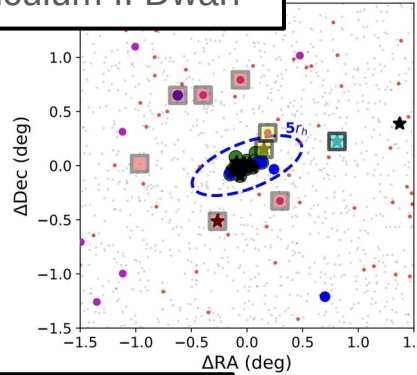
Do, Chiti et al. [2604.13374]

Metallicity-Sensitive Imaging for Dark Matter Science

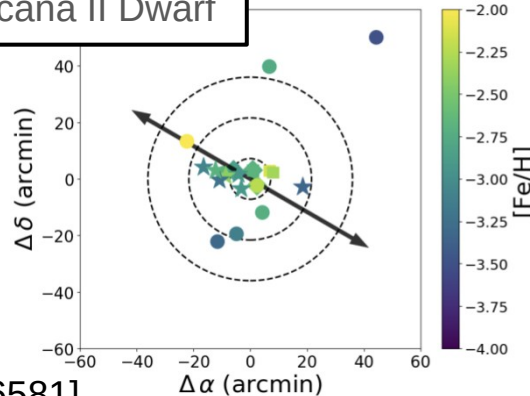
Pictor II Dwarf



Reticulum II Dwarf



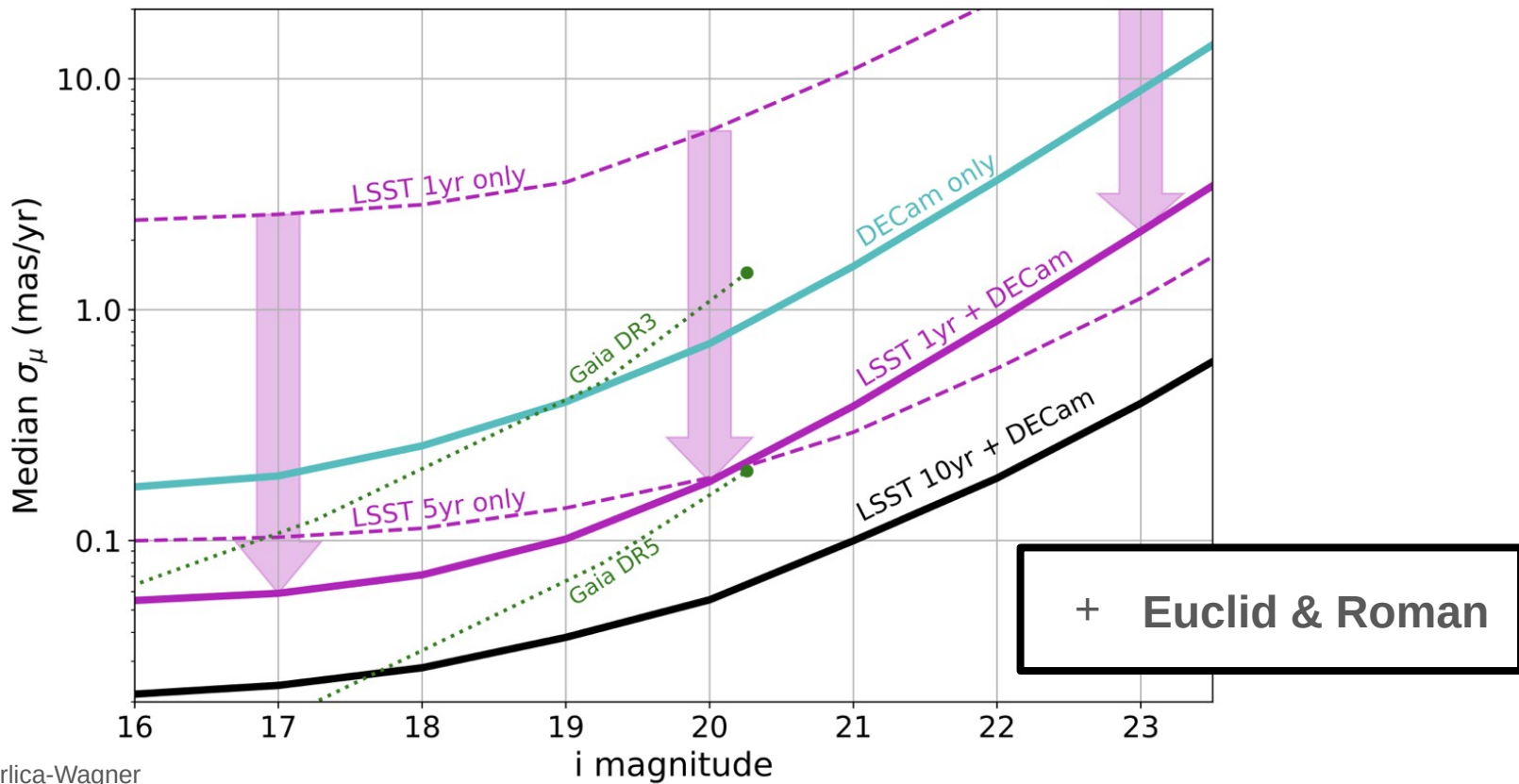
Tucana II Dwarf



- Extent of dark matter halos around ultra-faint dwarf galaxies.
- Signatures of tidal disruption/dynamical disequilibria
- Halos around dwarfs as an indications of early dwarf-dwarf mergers

& Proper Motion

Metallicity-Sensitive Imaging for Dark Matter Science



Gaining spectral information while retaining transient science: offset broad bands (Jeff Newman, U. Pittsburgh)

- Basic scaling of photometric redshift errors:

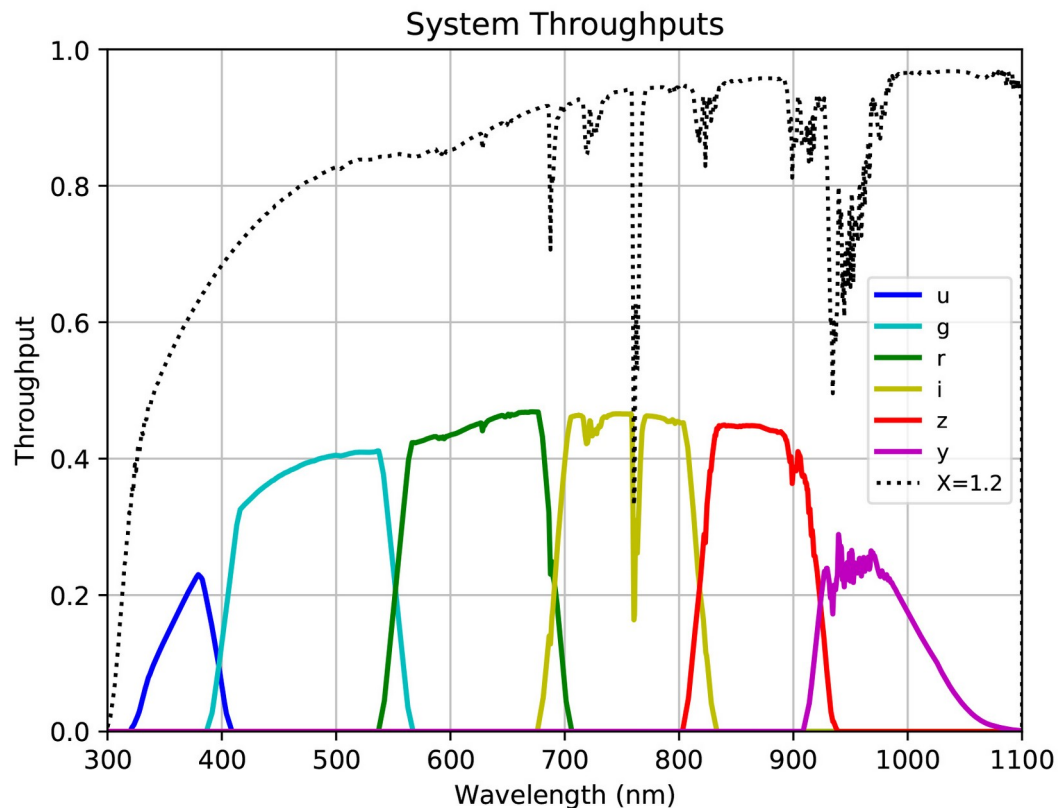
$$\Delta z \sim \frac{\Delta \lambda}{\lambda} \propto R^{-1/2} \times N_{\text{features}}^{-1/2} \times (\epsilon A t_{\text{total}})^{1/2}$$

where R is the spectral resolution, N_{features} is the number of features used to determine a redshift, ϵ is efficiency/throughput, A is collecting area, and t_{total} is total survey time

- Result: if double the resolution **AND ALSO** run for 10 more years, LSST photo-z errors would get reduced by a factor of 2
- Likely more interesting to do this over limited area (with longer time per field) so can get much smaller photo-z errors, though also over smaller volume

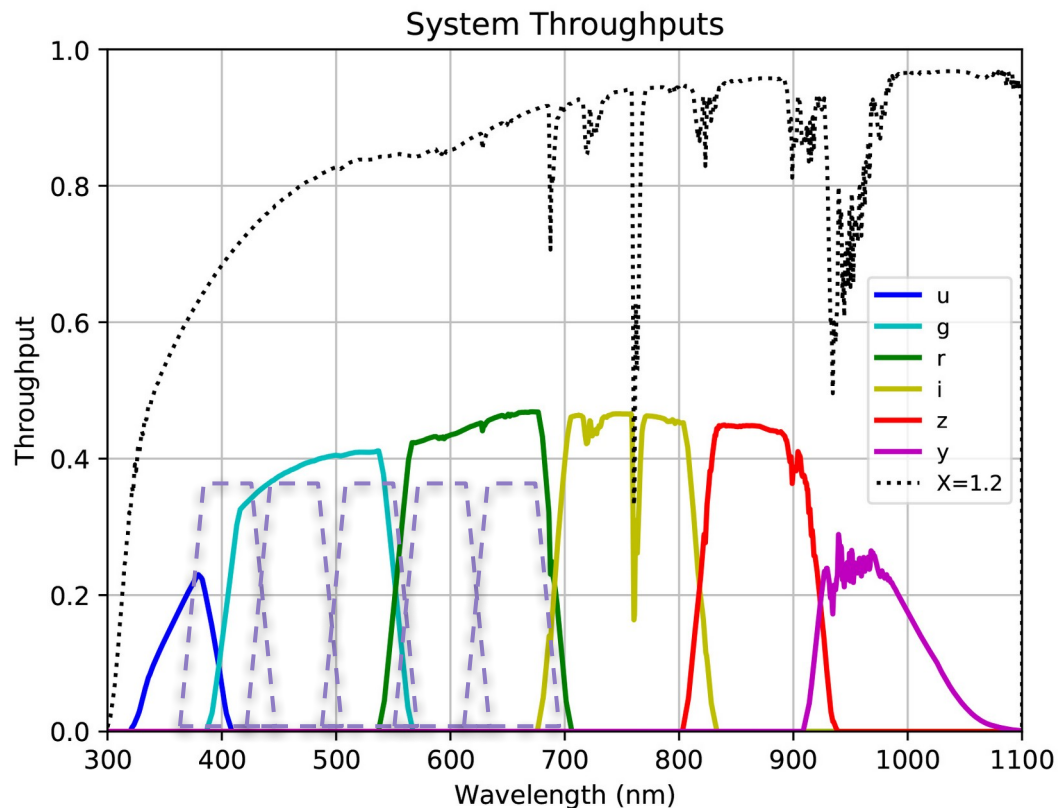
Gaining spectral information while retaining transient science: offset broad bands (Jeff Newman, U. Pittsburgh)

- If we start observing medium bands in 2036, can gain spectral resolution... but depth of imaging takes a hit
- Reduces sensitivity for finding transients and measuring light curves
- $\sim\sqrt{2}$ decrease in S/N if medium bands cover half of a broad band
- No gain for photo-z's when medium bands don't cover the wavelength of a spectral feature



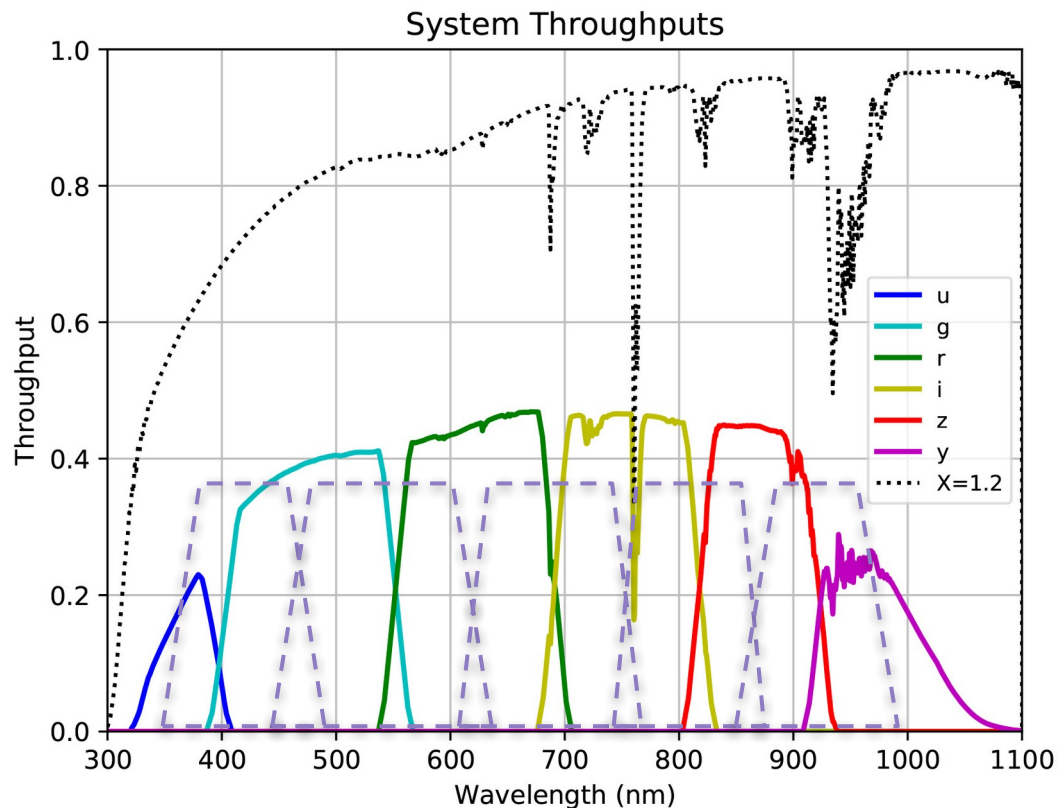
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Gaining spectral information while retaining transient science: offset broad bands (Jeff Newman, U. Pittsburgh)

- Alternatively, can use a set of broad bands that are offset in wavelength from the original set
- Provide additional spectral information in combination with *ugrizy*
- Retain full sensitivity to transients
- Hard to do at extreme wavelength ends though



Offset broad bands would have similar photo-z impact as medium bands, **BUT** still retain full transient sensitivity

- Empirical test: calculate random forest photo-z's from the 12 COSMOS2020 medium bands (centered at 427-827 nm), separately and in 'broad' combinations of 2, with $\sqrt{2}$ more noise added for medium-band measurements
- Photo-z errors are big because of limited wavelength coverage, noise in COSMOS2020 and large noise added ($\sigma=0.075$ for broad bands/10 years)

Scenario	Robust scatter (NMAD)	Outlier rate
6 broad bands (10 years total)	0.111	30%
6 broad bands (20y total)	0.096	28%
11 broad bands (with offset bands, 20y total)	0.076	26%
12 medium bands (20y total)	0.107	31%
hybrid: 6 medium bands in blue, plus 6 broad bands (20y total)	0.081	26%

A Truly Global Rubin

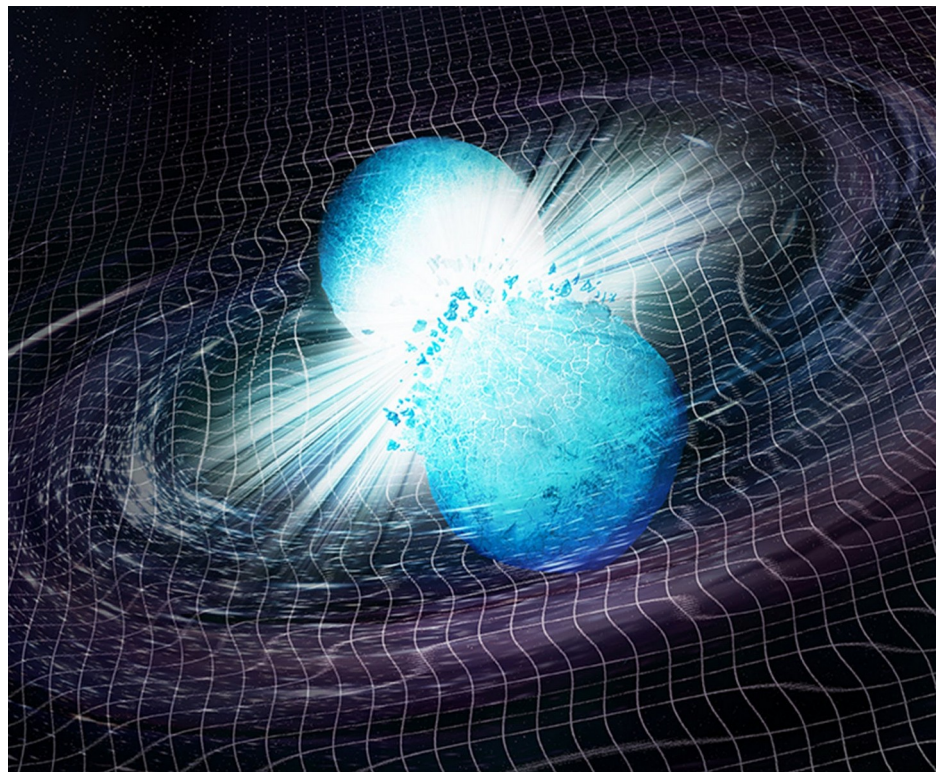
Michael Wood-Vasey

- Science:
 - See Saurabh Jha's talk.
 - Transients, full coverage for high-quality SNe LCs, quadrant filters (full, offset, medium).
- Community:
 - What does a Truly Global International Rubin Community Look Like?
 - How does the US-centered community switch to truly believe that the best science comes from everyone in the world asking questions and to act that way as a primary?
 - What would a series of Rubin2036@{Africa,Asia,Europe,India,LATAM,Pacific,...} look like?
- Data:
 - How can we develop cheaper ways to (process, store, serve) data+analysis?
 - Just as we have industrialized production of detectors, how do we achieve the same in software, pipelines, and curation?

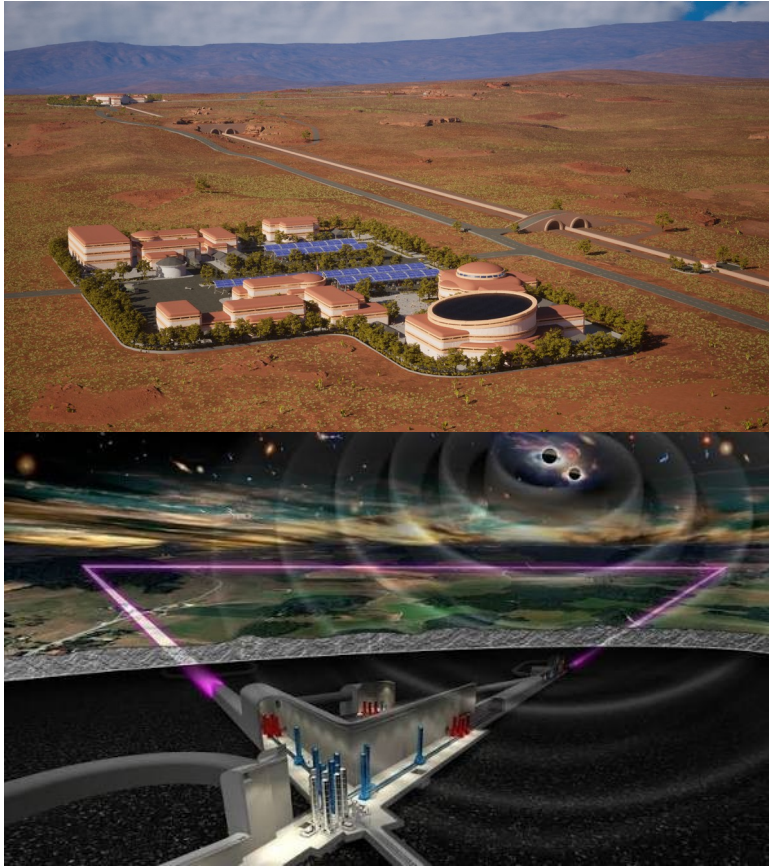
GW Cosmology and Astrophysics in 2036 with Rubin Observatory

Sean Patrick MacBride, University of Zurich

- **Bright siren detection:** Mergers at higher redshift from:
 - Binary neutron stars detected by Cosmic Explorer and/or Einstein Telescope.
 - Massive black hole binaries detected by LISA.
- **Pre-merger monitoring:** enabled by LISA detection of massive black hole binaries.
- **Host galaxy environments:** Correlating GW source properties with their host galaxy properties to reduce systematics for dark siren measurements.
- **Modified GW propagation:** High redshift sources identified from Cosmic Explorer and/or Einstein Telescope signals.

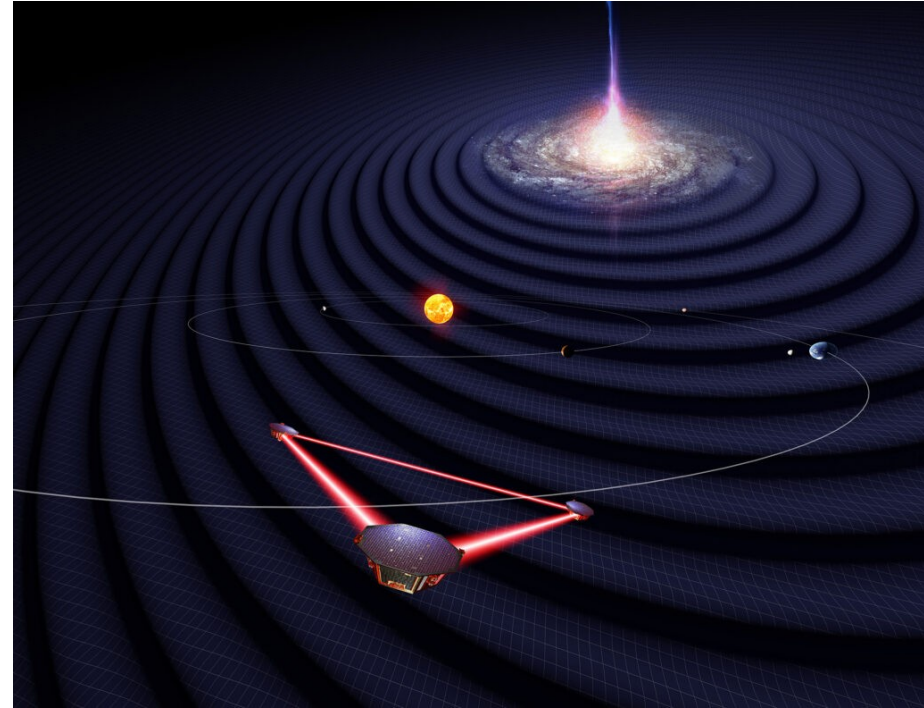


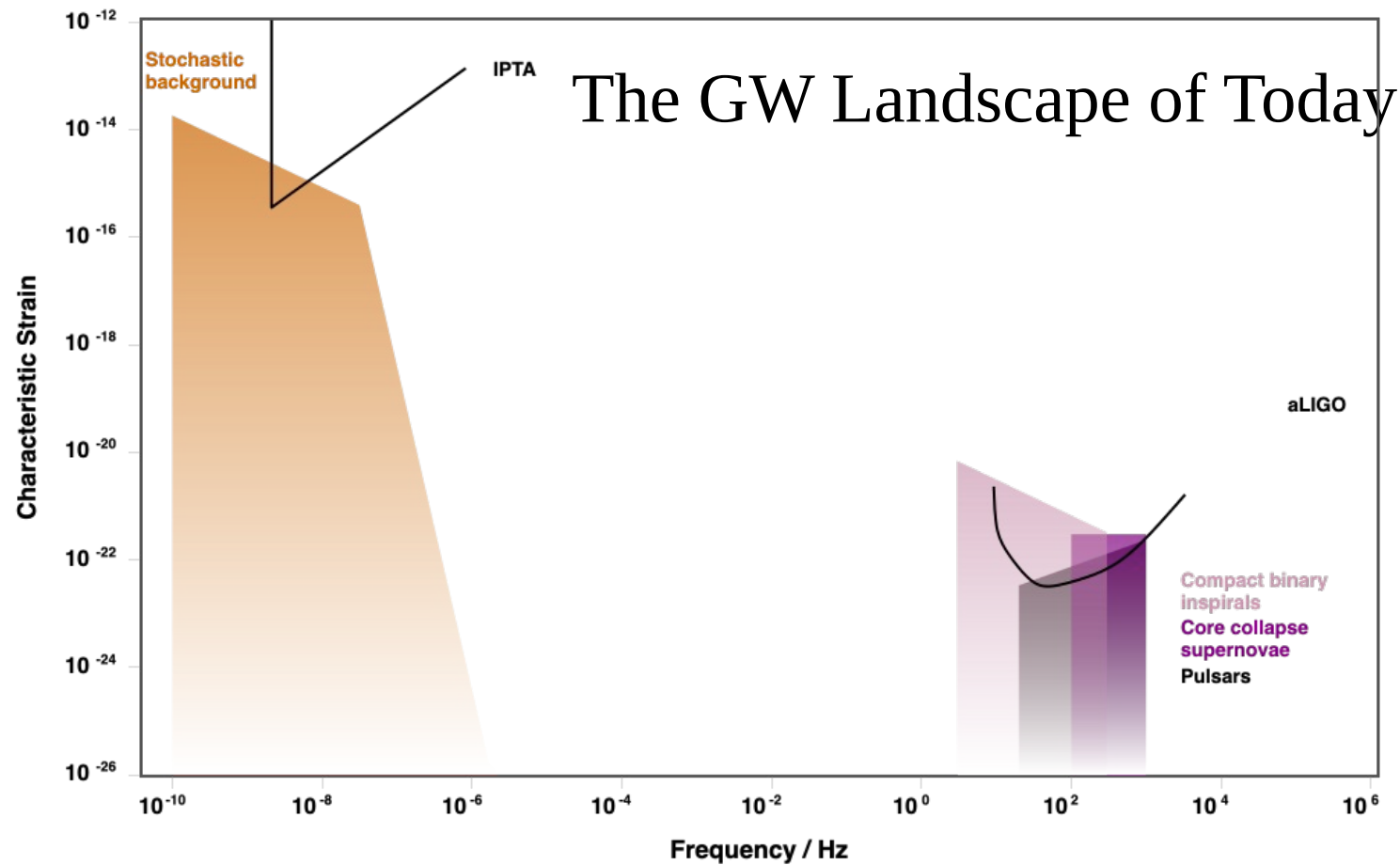
Cosmic Explorer - Ready in 2040

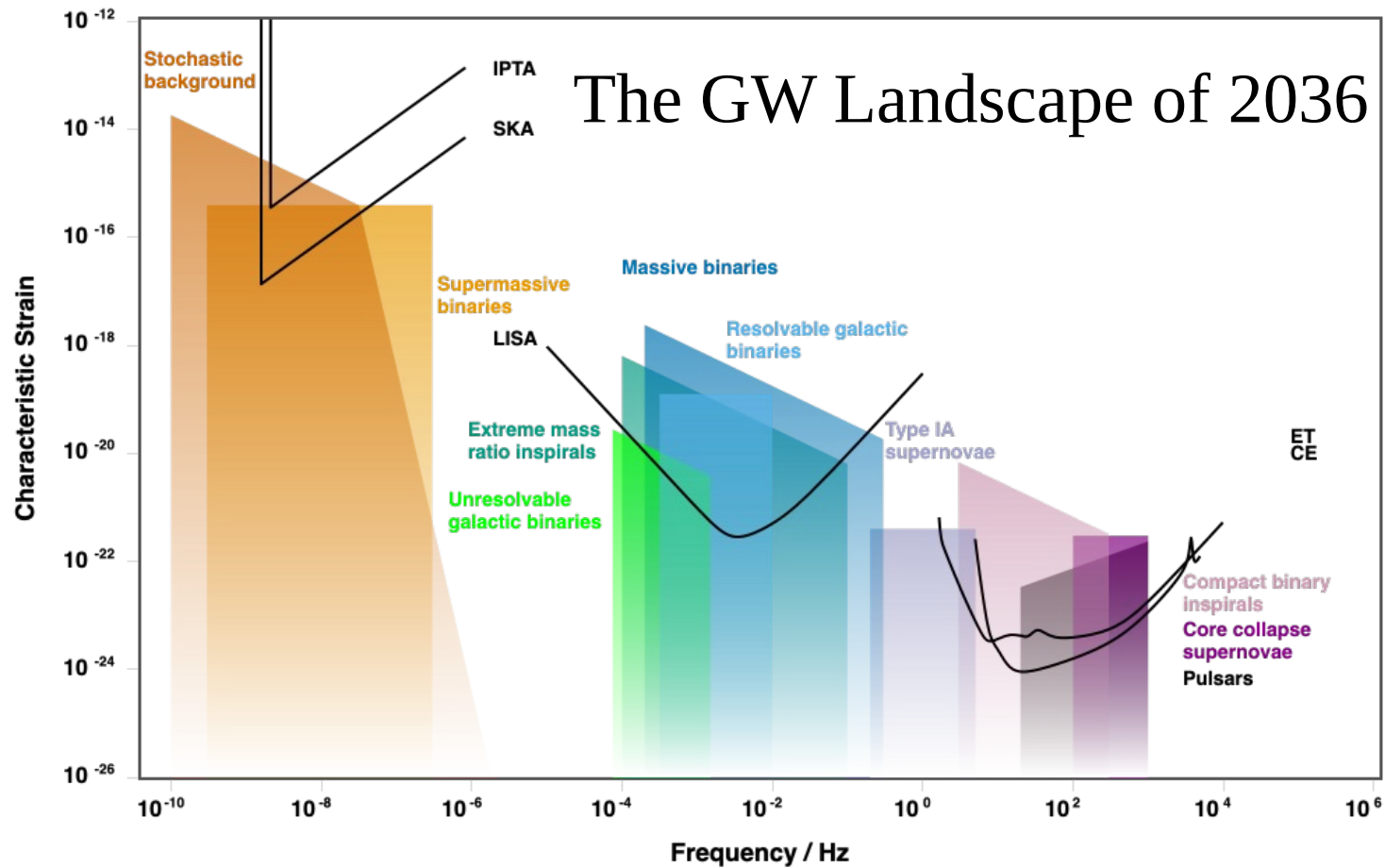


Einstein Telescope - Ready in 2035

Laser Interferometer Space Antenna - Launch in 2035







Rubin FOV is Ideal for Discovery

Is the Rubin FOV effective for identifying these transients?

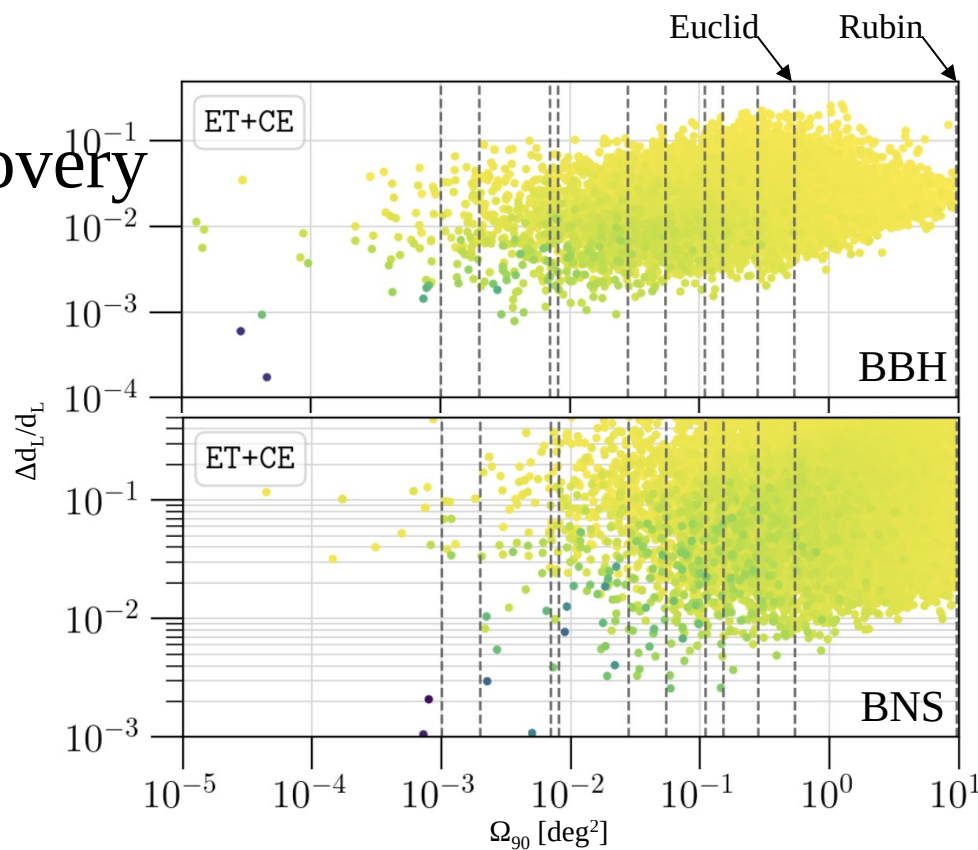
- Rubin FOV is highly compatible with well-localized BNS and BBH mergers from Cosmic Explorer and/or Einstein Telescope.

Could Rubin detect massive black hole binaries from LISA?

- Detectable by Rubin out to $z \sim 4$, though horizon of detection is highly dependent on AGN disk model.

Could Rubin detect binary neutron stars detected by CE + ET?

- Rubin can detect the electromagnetic counterparts to ~ 100 binary neutron stars per year with Einstein Telescope alone.
- Cosmic explorer will enable more detections and better localization.



The 90% localization area and fractional luminosity distance of forecasted Cosmic Explorer BBH and BNS mergers.

Can Rubin Detect Any of the Electromagnetic Counterparts?

Is the Rubin FOV effective for identifying these transients?

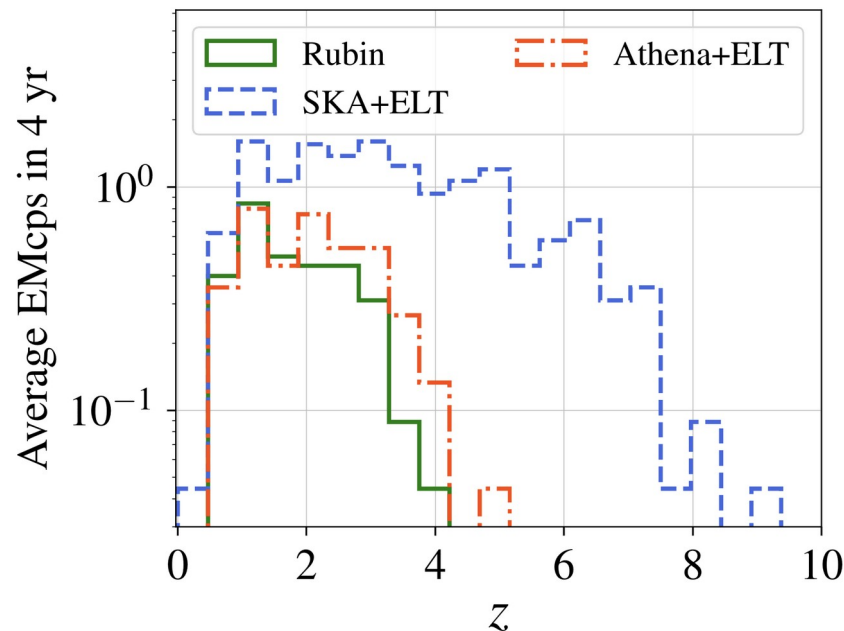
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Average number of electromagnetic counterparts detected in 4 years by different instruments, associated with LISA massive black hole binary detection, using the Q3d emission model.

Can Rubin Detect Any of the Electromagnetic Counterparts?

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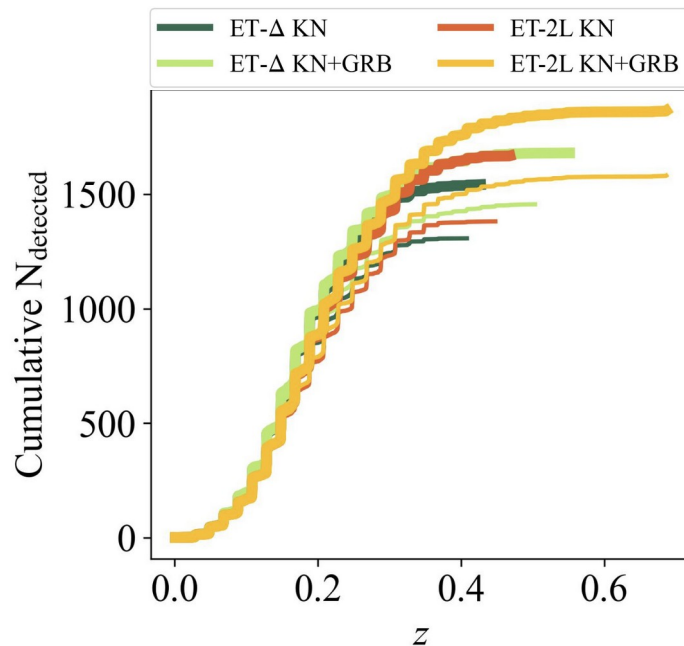
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Cumulative number of KN and KN+GRB counterpart detections by Rubin Observatory from Einstein Telescope observations over ten years of operations. Thick lines indicate the 1200s exposure time, and the thin lines indicate 600s exposures [5].

Medium-band target selection of Lyman alpha emitters

Haruki Ebina (UC Berkeley)

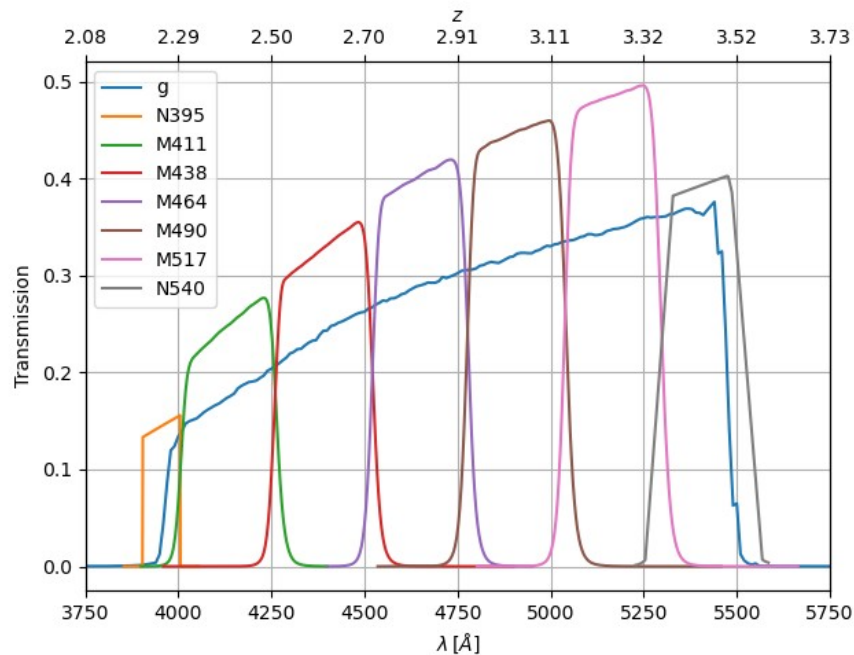
Medium bands can be used to select Lyman alpha emitters (LAEs) at $z > 2$

We have confirmed this using IBIS
(Intermediate Band Imaging Survey) + DESI

- See Raichoor et al. (in prep) for work with Subaru Suprime MBs + DESI

IBIS covers wide fields ($\sim 3000 \text{deg}^2$) to $m=25$,
deep fields to $m=25.5$

See Hironao Miyatake's talk tomorrow for
HSC-Niji, a complementary MB survey to IBIS

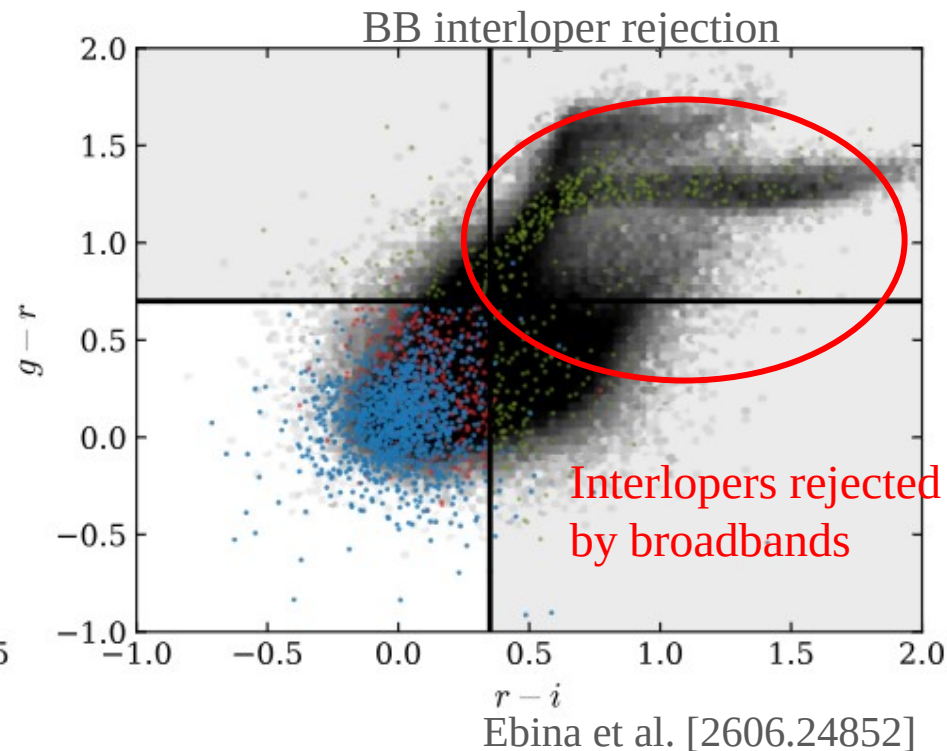
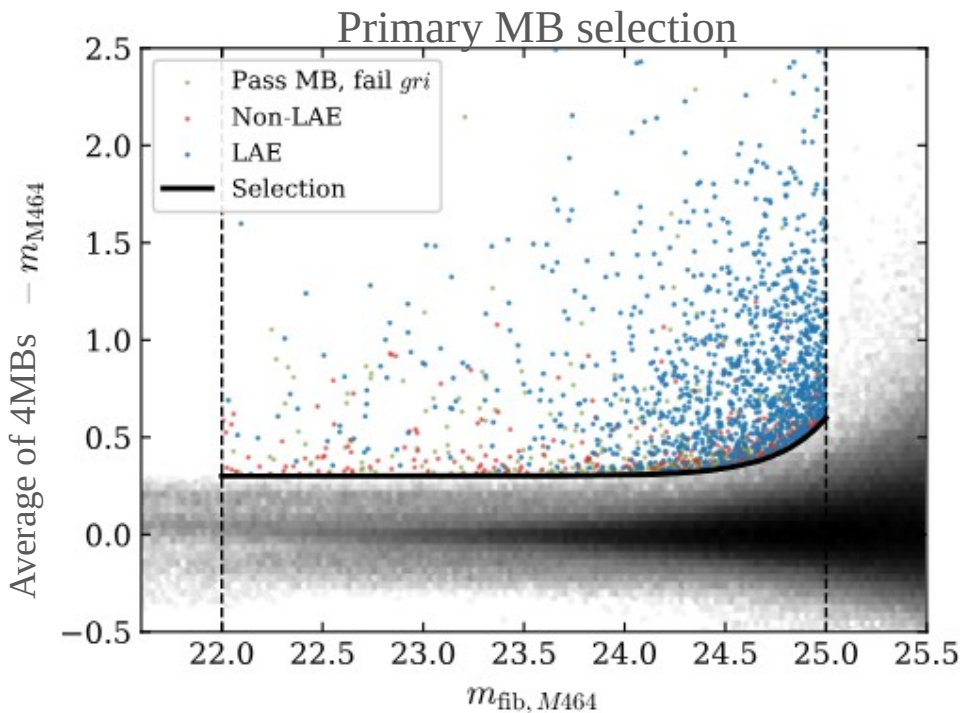


Credits: A. Raichoor

Medium-band target selection of Lyman alpha emitters

Haruki Ebina (UC Berkeley)

2/4



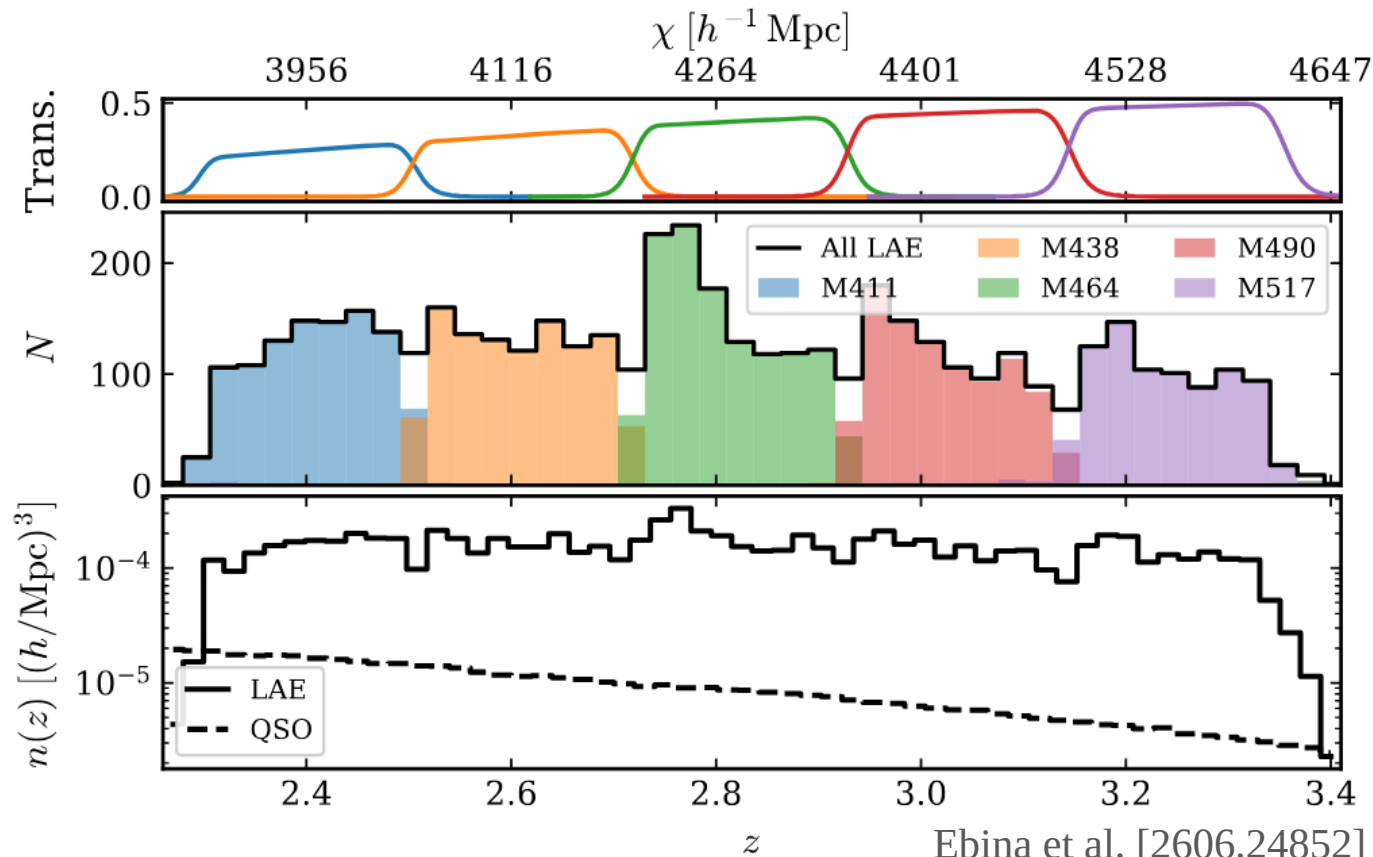
In total, 650/deg² LAEs to the shallow (wide-field) depth

75% total purity (target selection purity + spectroscopic efficiency)

Medium-band target selection of Lyman alpha emitters

Haruki Ebina (UC Berkeley)

3/4



The target galaxies match the distribution of MBs, as expected

Medium-band target selection of Lyman alpha emitters

Haruki Ebina (UC Berkeley)

LAEs are planned for observation in DESI Run 2 (2029-2035), a pathfinder for high- z LSS cosmology

Expanding this to deeper imaging, wider redshift range and wider area can provide dramatically more galaxies, potentially leading to a spectroscopic Stage-V survey (SpecS5) – see Claire Poppet's talk tomorrow

There are a lot of science cases for high- z cosmology! See e.g. Schlegel et al. (arxiv:2209.03585), Ferraro et al. (arxiv:2203.07506) and Besuner et al. (arxiv:2503.07923)

Automated Anomaly Detection and Transient Classification

[John F. Suárez-Pérez](#) (Tecnológico de Monterrey)

Challenge

- ☐ Rubin will capture light curves for billion sources and produce **millions of transient candidates per night** ^[1]
- ☐ It will generate a live alert stream for the entire 10-year survey
- ☐ Visual inspection is impossible at this scale where **automation is mandatory**
- ☐ Standard pipeline flags miss a **significant fraction** of problematic data (Data Quality)

Solution

Learn what "normal" data looks like directly from the data itself – no labeled training data required (current brokers). Outliers become visible as isolated structures in the embedding ^[2].

Component	Method	Purpose
Dimensionality Reduction	Unsupervised ML	Project high-dimensional spectra/images into 2D space
Outlier Detection	Clustering	Identify isolated points and small groups in the embedding
Visual Inspection	Targeted review	Confirm anomalies and diagnose root causes

[1] [The automation of optical transient discovery and classification in Rubin-era time-domain astronomy](#)

[2] [Quality Assessment of Spectroscopic Data Reduction Pipelines using Artificial Intelligence](#)

Automated Anomaly Detection and Transient Classification

[John F. Suárez-Pérez](#) (Tecnológico de Monterrey)

Evidence from DESI Data Release 2 (58M Spectra) [1]

Methodology:

- Processed **58M spectra** from **14k tiles**
- Applied **Unsupervised ML** (dimensionality reduction) + **(FoF, kNN)** (clustering)

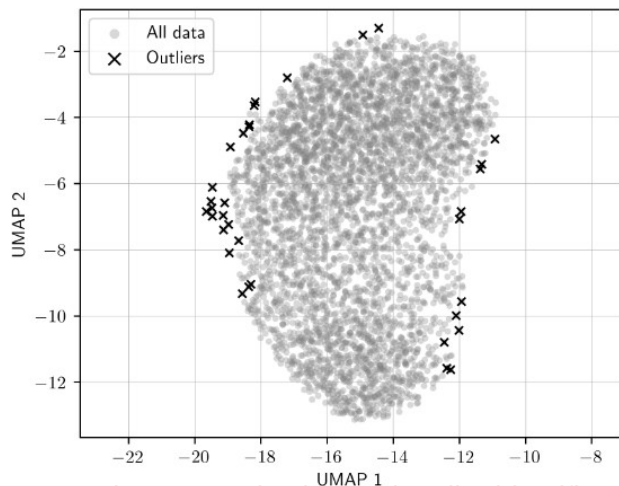


Fig1. 2D projection and outlier identification

Metric	Result
Mean outlier rate	~1.96% per tile
Purity (visually confirmed anomalies)	From a sample of 391 outlier candidates, 66.8% (dark program) / 63.5% (bright program)
Flagged by standard pipeline (ZWARN≠0)	Only 4.1%

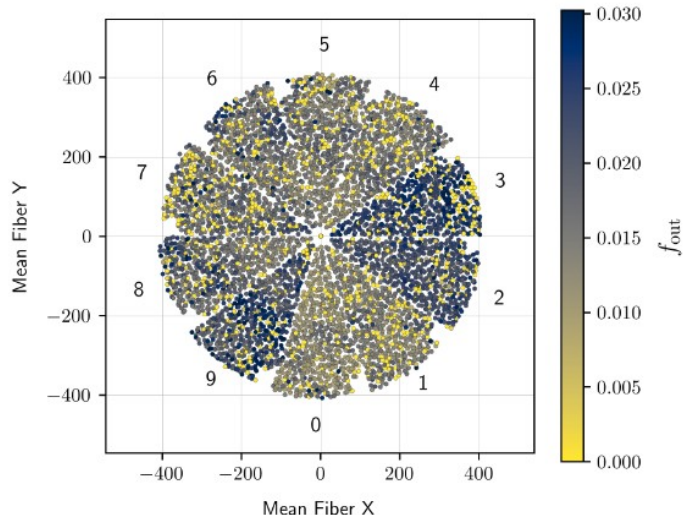


Fig2. Outlier fraction distribution on the focal plane

Automated Anomaly Detection and Transient Classification

[John F. Suárez-Pérez](#) (Tecnológico de Monterrey)

Adapting to Rubin: From Spectra to Light Curves

Aspect	DESI (Spectra)	Rubin (Light Curves / Images)
Data type	1D flux vs. wavelength	Time series
Dimensionality	~6,000 wavelength bins	100-1000 epochs per object
Representation	Direct vector input	Feature extraction required
Anomalies	Arm-join discontinuities	Bad epochs, outliers, calibration issues

- Brokers classify what they know (supernovae, AGN, etc.)
- To detect what brokers **don't** know (novel phenomena, artifacts)
- To improve broker purity by flagging data quality issues

Benefits to Rubin

- ☐ **Data Quality:** Catch anomalies standard pipelines miss → better cosmology
- ☐ **Efficiency:** Reduce human inspection from millions to thousands of candidates
- ☐ **Discoveries:** Identify rare transients and unexpected phenomena
- ☐ **Scalability:** Proven on 58M spectra → ready for Rubin's 10B sources

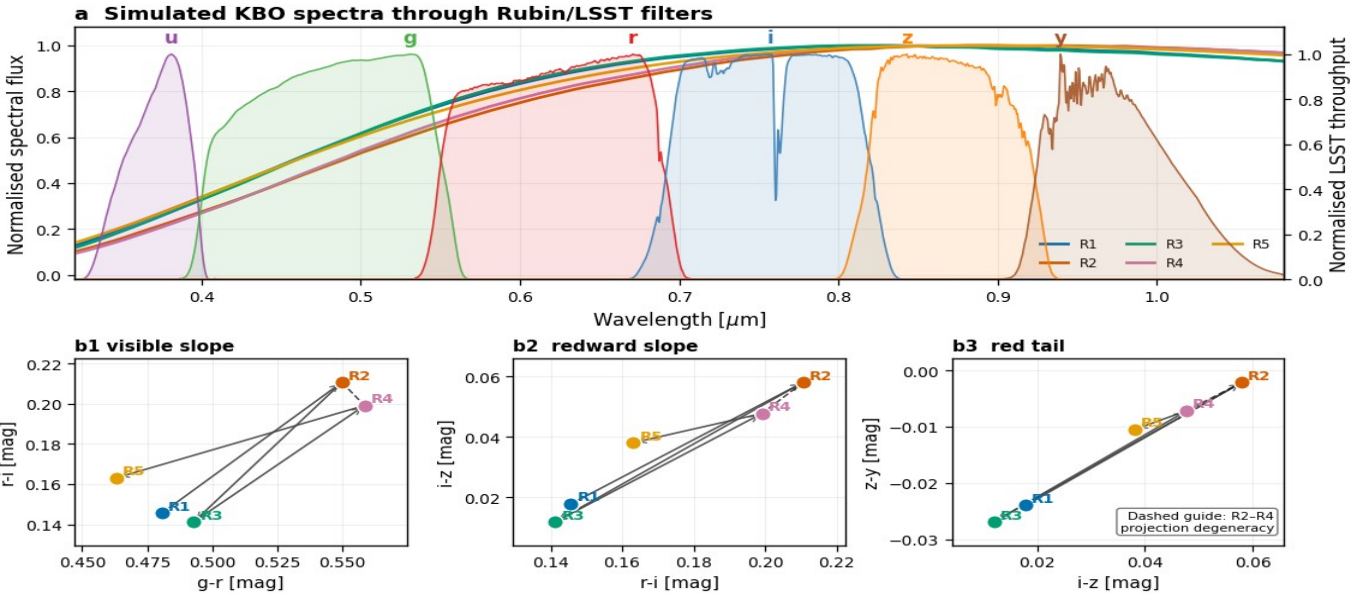
**Broadband Rubin
Photometry as a
First-Stage Physical
Discriminator for Outer Solar
System Targets**

Becky Long - Nigel J. Mason -
Andjelka Kovačević



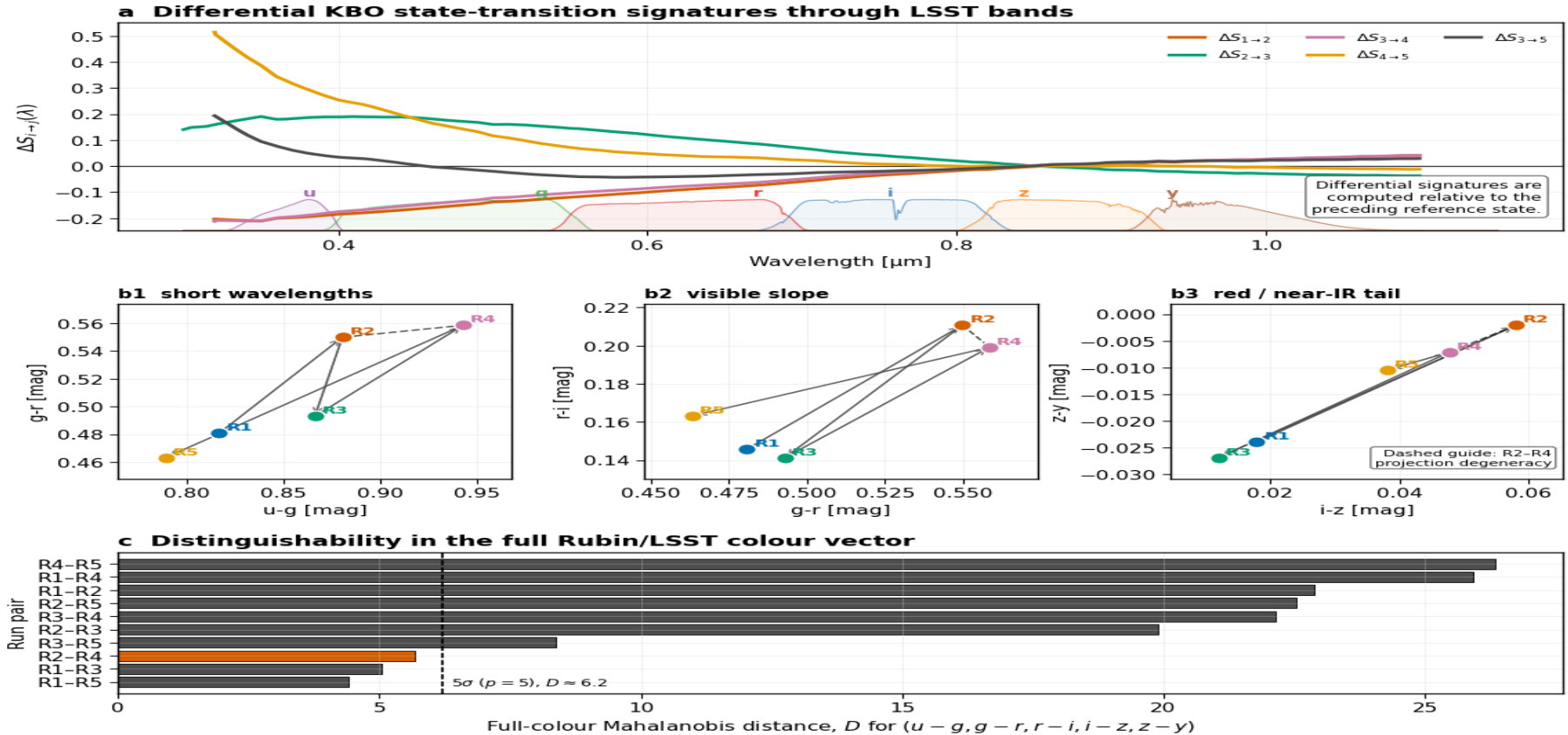
KBO states in Rubin colour space from PSG simulations

Run	Spectral-analogue composition	Physical state	Expected colour / science
R1	100% achondrite	Inactive rocky baseline	Featureless photometric floor
R2	40% organic + 60% achondrite	Organic-rich irradiated	Strong reddening in g-r and r-i
R3	30% ice analogue + 70% achondrite	Ice-rich surface	Reflective; greyer; near baseline
R4	40% achondrite + 30% ice + 30% organic	Mixed ice-organic	Blended red/grey signature
R5*	R4 + activity component	Exploratory weak activity	Grey/chromatic displacement



Graph from Long et al. 2026 / Kovačević et al. 2026, IAU 404

Differential signatures + Colour planes + Mahalanobis distance



The Rubin 2036 decision we can support

Can a blue-visible medium band add enough physical information to justify its observing cost?

1 GOAL

Test separation of organic-rich, ice-rich, mixed and weakly active KBO states beyond ugrizy.

2 WHY RUBIN / WHY NOW

Unique étendue + repeated multiband imaging + decade baseline; filter development needs ~2-3 years.

3 SKEPTIC'S OBJECTION

Colours are degenerate and spectroscopy exists. Answer: physical triage reduces ambiguity at population scale.

4 WHAT RUBIN NEEDS

Test a candidate near 450 nm; optimise central wavelength and FWHM against throughput and cadence cost.

5 REQUIRED PRIORS

DR3: colour-error covariance and completeness. DR6: multi-epoch chromatic and activity baselines.

6 SUCCESS METRIC

Gain in minimum Mahalanobis D versus ugrizy at fixed observing cost; target at least 25% for the hardest pairs.

7 CONNECTIONS

Rubin Solar System processing and brokers to Gemini/GOATS, JWST and future ELT confirmation.

8 NEXT STEP + OWNERS

Run wavelength x width x exposure grid: Long, Mason, Kovačević + Rubin filter, Solar System and strategy teams.

AI/ML

Accelerates the PSG design grid and next-observation optimisation; it is not the science classifier.

CHILE

Co-design with Chilean Rubin/NOIR Lab researchers; connect Chile-based follow-up and training exchange.

DELIVERABLE | Quantitative filter + physical-priority layer - not classifier

References

- Kovačević A. B., Mason N. J., Ćiprijanović A., Long B. et al. (2026), IAU Symposium 404 Proceedings (submitted)
- Long B., Mason N. J., Kovačević A. (2026), EPSC2026-84, doi:10.5194/epsc2026-84
- Kovačević A., Mason N. J., Ćiprijanović A. (2025), *Frontiers in Astronomy and Space Sciences*, 12, 1 594 485
- Schwamb M. E. et al. (2023), *ApJS* 266, 22 · Kofman V. et al. (2024), *PSJ* 5, 197 · Villanueva G. L. et al. (2022), *PSG Fundamentals*
- Soraisam et al. (2026), *GOATS*, arXiv:2606.28645.
- Xkcd science comics, Vera Rubin Observatory, xkcd: Vera Rubin Observatory

Thank you!

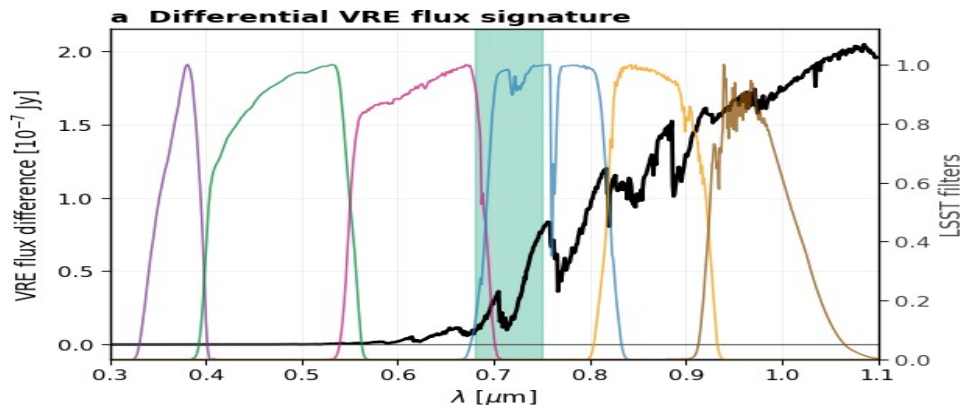
VERA RUBIN OBSERVATORY DISCOVERIES
BINGO CARD

BLACK HOLE TIDALLY DISRUPTING A STAR	PLANET NINE	REMNANTS OF A FORMER MILKY WAY COMPANION GALAXY	10,000 NEW TRANS- NEPTUNIAN OBJECTS	OBJECT ON INTERSTELLAR TRAJECTORY
STAR BEING ECLIPSED BY WEIRD OBJECT	SECRET MILITARY SATELLITE	KUIPER BELT OBJECT THAT'S A SURPRISING COLOR	MARS- SIZED TNO	LOCAL GROUP SUPERNOVA
SPURIOUS SIGNAL FROM OBSERVATORY KITCHEN	GLITCH IN ALERT SYSTEM SPAMS ALL ASTRONOMERS	SPACE (FREE)	EARTH- SIZED TNO	DEFUNCT SPACECRAFT
SURPRISING NUMBER OF SMALL GALAXIES	NEUTRON STAR MERGER	NEW SUPERNOVA CONSTRAINTS ON DARK ENERGY	NEW DWARF PLANET BIGGER THAN PLUTO/ERIS	LOST HISTORICAL COMET
5 MILLION ASTERIODS AND COMETS	50,000 NEAR- EARTH ASTERIODS	OBJECT LIKELY TO IMPACT EARTH	VERA RUBIN HERSELF (FAKED DEATH, STOLE SPACESHIP)	MANY VERY SMALL DOTS

Can one 710 nm band turn a weak red-edge signal into a decision-grade Rubin measurement?

A physics-based filter trade study—not a photometric claim of biosignature confirmation

PSG forward models locate the information gap near 710 nm



Physical design space

Central wavelength ≈ 710 nm

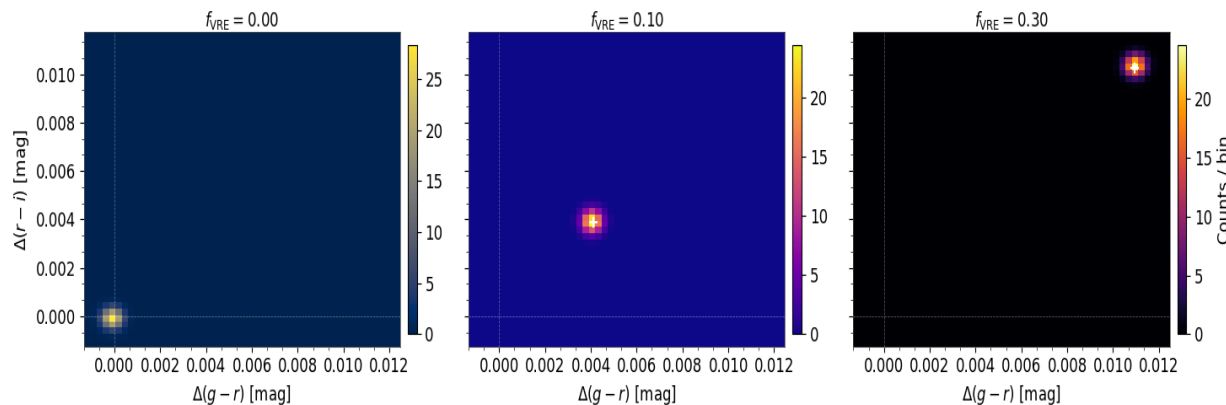
Width 20–40 nm

Cadence coordinated r-710-i

Systematics DR3/DR6 covariance floor

Start from physical spectra, then ask whether one added band reduces the required stacking.

The signal is coherent—but sub-threshold in ugrizy



649

PSG spectra

0.25σ

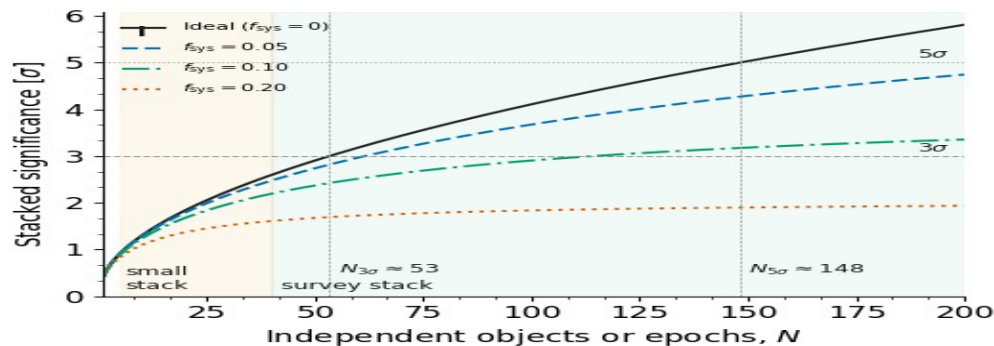
$f_{VRE} = 0.1$

0.63σ

$f_{VRE} = 0.3$

What we already have: a forward model, a nuisance-aware colour statistic, and a quantitative baseline for filter optimisation.

Stacking helps; target geometry sets feasibility



PAPER RESULT

$$N_{3\sigma} \approx 53 \cdot N_{5\sigma} \approx 148$$

Strongest intrinsic simulation: fVRE = 0.3

Optimistic 5-scale-height H_2 screen

System	3σ at 0.01	3σ at 0.001
GJ 1214 b	≈ 338	$\approx 3-4$
TOI-270 d	$\approx 20,100$	≈ 201
K2-18 b	$\approx 65,100$	≈ 651
LHS 1140 b*	$\approx 15,900^*$	$\approx 159^*$

per-transit colour precision (mag)

At 0.001 mag, GJ 1214 b reaches a nominal 3σ in $\approx 3-4$ transits.

A decision-ready roadmap, with confirmation outside Rubin

GOAL

Quantify gain from a ~ 710 nm band

WHY NOW

Rubin A $\cdot\Omega$ ·T; filter lead >2–3 yr

NEED

One filter + r-710-i + software

PRIORS

DR3 covariance; DR6 false-positive baselines

METRIC

$N_3\sigma/N_5\sigma$, false positives, candidates/hour

OBJECTION

Non-unique unresolved colours

ANSWER

Physical triage, not confirmation

NEXT STEP

End-to-end star+planet+instrument grid

AI/ML accelerates the physical grid; Chilean co-leadership should shape filter design, data access and follow-up training.

Owners: Mason • Moore Kovačević • + Rubin filter/camera, calibration, stellar-variability and survey-strategy contributors

References and claim boundary

Primary sources

Kovačević, A. B., Mason, N. J. & Moore, M. (2026)
Prototype Coherence-Based Biosignature Searches with Vera C.
Rubin Observatory / LSST. Rubin 2036 Group C pre-work.

Kovačević, A. B., Mason, N. J. & Moore, M. (2026)
EPSC-DPS2026-763. doi:10.5194/epsc-dps2026-763

Current result

A coherent ugrizy displacement and idealised
stacking baseline.

Next test

Whether ~710 nm reduces N after stellar dilution
and Rubin systematics.

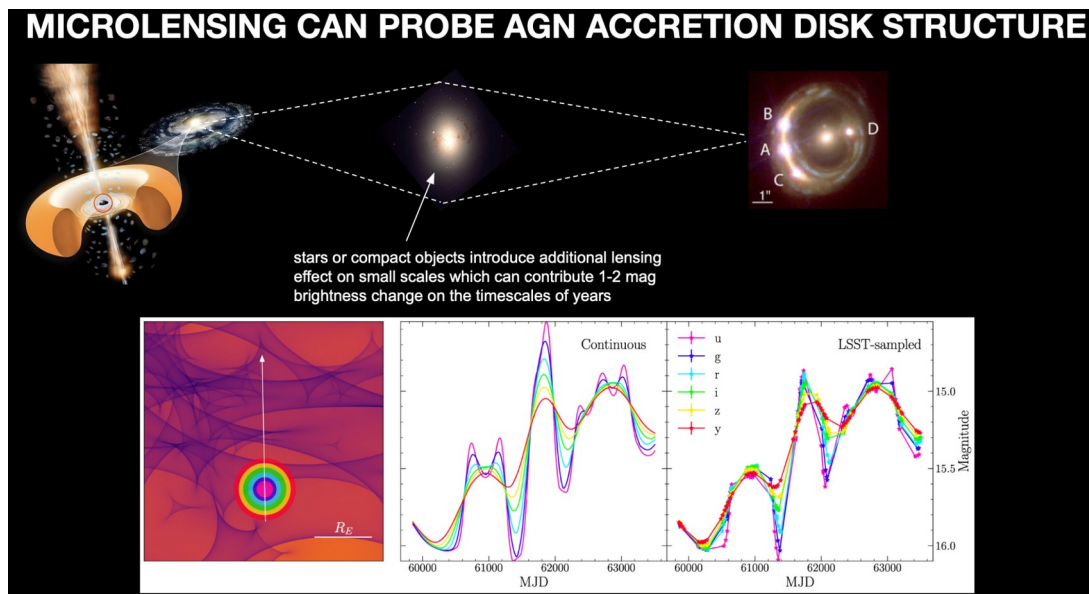
Confirmation

Requires independent spectroscopy.

Supporting methods: Villanueva et al. (2022), PSG fundamentals · Pont, Zucker & Queloz (2006), MNRAS 373, 231 · Kovačević, Mason & Čiprijanović (2025), Frontiers 12, 1594485

My strong lensing crystal ball prediction for Rubin 2036

- Lensed quasars in multi-band monitoring - a unique probe of AGN accretion size and micro-lensing! Good cadence in multi-band is key - unclear whether achievable with LSST Y10.
- Lensed SNe - Will be a struggle throughout Y10. They are hard to find and require immediate follow up for depth and cadence to measure time delays. Follow-up limited.
- Alternative cadence on Roman footprint of $\sim 2000 \text{ deg}^2$ (5x cadence and 2x exposure time relative to LSST to compromise smaller area).



Exoplar

[Exoplanet L](#)

The Rubin Observatory

When LSST ends

[Open Preview](#)

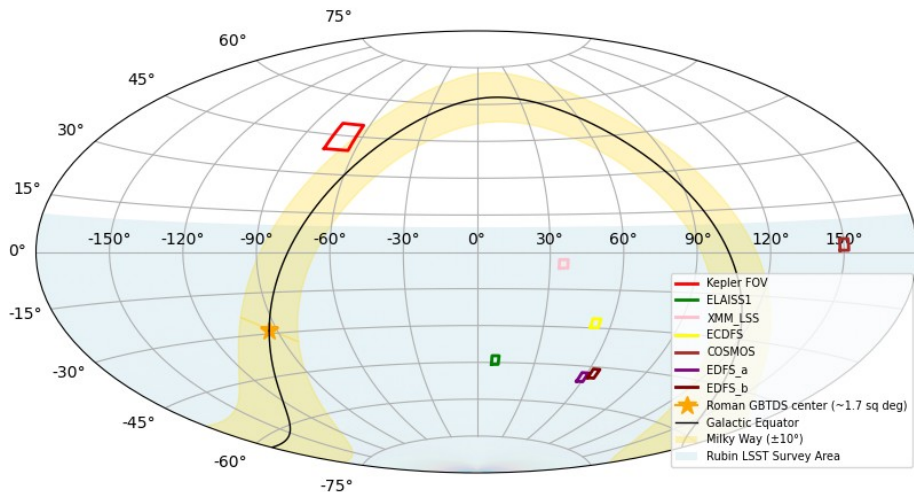
Transiting Planets

The Exoplanet Gaps

(Don't we have a lot of good survey data already?)

Kepler's fixed field covered only $\sim 0.25\%$ of the sky, leaving the MW's broader stellar population essentially unmapped for planet demographics.

Kepler Field, Rubin LSST, Roman GBTDS and Milky Way Band



On the other hand, TESS covered most of the sky — but only its brightest, nearest stars ($M \leq 12$). As a result, The Milky Way's exoplanet population lives mostly beyond its reach.

One filter to rule them all (and also cadence optimization)

Alex Malz (STScI)

Consider spatially-varying filter (group C)

- Narrow-, medium-, broad-band, or comb coverage; continuous or segmented

- Eliminates filter exchanger liability and time overhead

- Build differential spectra instead of co-adds as source observed across filter

- Bayesian (wavelength-)pixel-level processing within reach already

Aside: Consider information-theoretic cadence optimization approach (group B)

- All metrics in terms of information about science goal

- Separates human decision of science prioritization from metrics'

sensitivity

Kilonovae detections leveraging Rubin's synergies with 3G GW detectors

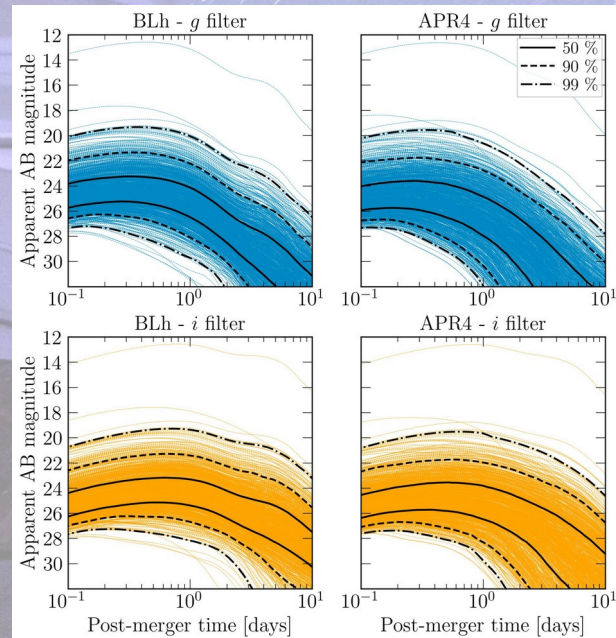
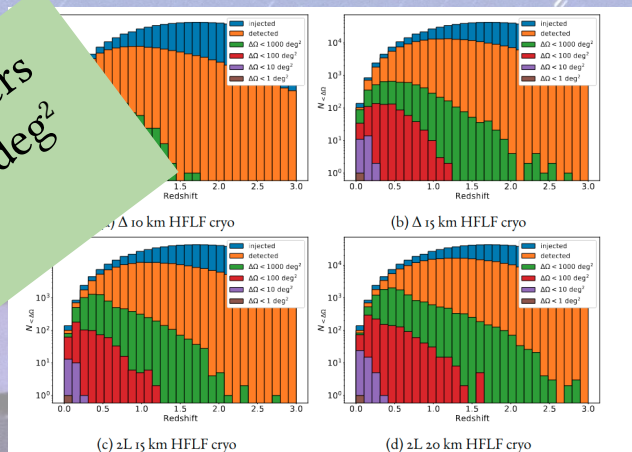
Nandini Hazra (NCBJ Warsaw)



$O(100)$ BNS mergers
localized $<100 \text{ deg}^2$

Prospects for optical detections from binary neutron star mergers with the next-generation multi-messenger observatories

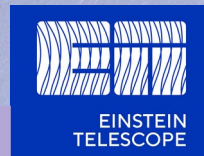
E. Loffredo^{*,1,3,2}, N. Hazra^{**1,3,2}, U. Dupletsa^{***1,2}, M. Branchesi^{1,2}, S. Ronchini^{4,2}, F. Santoliquido^{1,2}, A. Perego^{5,6}, B. Banerjee^{1,2}, S. Bisero⁷, G. Ricigliano⁸, S. Vergani⁷, I. Andreoni^{9,10,11,12}, M. Cantiello³, J. Harms^{1,2}, M. Mapelli¹³, G. Oganesyan^{1,2}.



Loffredo, NH, Dupletsa, Branchesi et al.



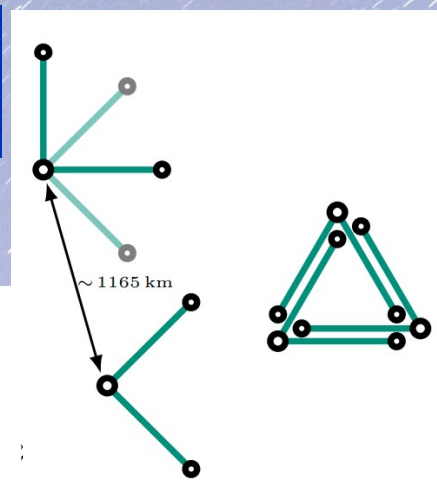
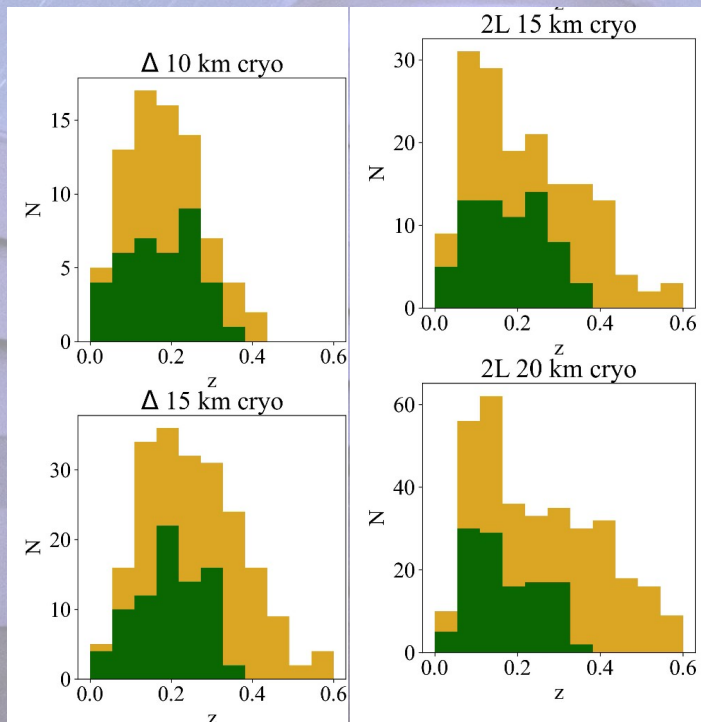
Kilonovae with Rubin and Einstein Telescope



Site selection in 2026!
Operational mid-2030s

Δ -10 km, Δ -15 km,
2L-15 km, 2L-20 km

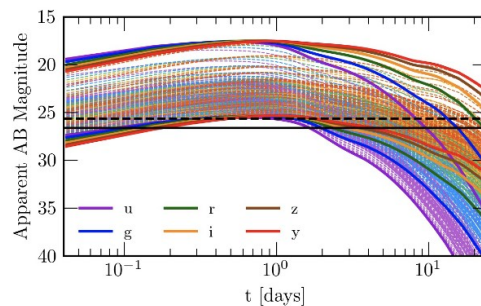
600 s ToO, $g+i$ filters, mosaic
strategy
1800 s exposures: prohibitive



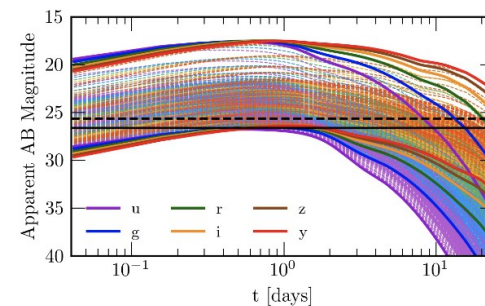
Kilonovae with Rubin and Einstein Telescope: detector design

Δ -10 km, Δ -15 km,
2L-15 km, 2L-20 km

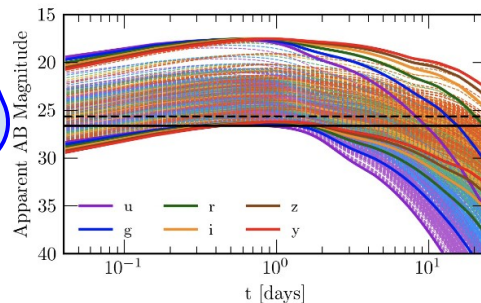
30-100 joint detections per
year ET-Rubin



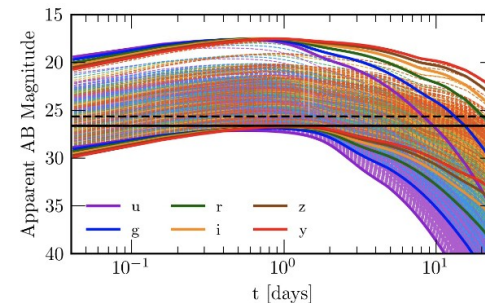
(a) Δ 10 km HFLF cryo



(b) Δ 15 km HFLF cryo



(c) 2L 15 km HFLF cryo



(d) 2L 20 km HFLF cryo

Einstein Telescope

detector design study



ET-Rubin synergy in multi-messenger astronomy is crucial to determining which ET configuration will be built

Science with the Einstein Telescope: a comparison of different designs

Marica Branchesi,^{1,2,*} Michele Maggiore,^{3,4,*} David Alonso,⁵
Charles Badger,⁶ Biswajit Sanyal,⁷ Freija Beirnaert,⁷
Enis Belgacem,^{3,4} Anurag Aggarwal,^{8,9} Guillaume Boileau,^{10,11}
Sohrab Borker,¹² David Brown,¹³ Man Leong Chan,¹⁴
Giulia Ciani,¹⁵ Yashir Hashim,^{16,17} Jerome Degallaix,¹⁸
Valeria D'Amico,¹⁹ Tim Dietrich,^{21,22}
Gabriele Franciolini,⁸
Boris Goncharov,^{1,2}
Nandini Hazra,^{1,2,27}
Kamil Janssens,^{10,11}
Yufeng Li,^{34,35}
Michele Mancarella,^{3,4,37,38}
Ana Martinovic,⁶ Andrea Maselli,^{1,2}
L. Miller,^{43,16,26} Chiranjib Mondal,²⁵
Harsh Narola,^{16,26} Micaela Oertel,⁴⁴
Costantino Pacilio,^{8,37,38} Cristiano Palomba,⁴⁵
Antonio Pasqualetti,⁴⁶ Albino Perego,^{47,48}
P  rigeois,^{39,40,41} Mauro Pieroni,^{49,50}
Annella Juliana Piccinni,⁵¹ Anna Puecher,^{16,26} Paola Puppo,⁴⁵
Angelo Ricciardone,^{52,39,40} Antonio Riotto,^{3,4} Samuele Ronchini,^{1,2}
Mairi Sakellariadou,⁶ Anuradha Samajdar,²¹
Filippo Santoliquido,^{39,40,41} B.S. Sathyaprakash,^{20,53,54}
Jessica Steinlechner,^{16,17} Sebastian Steinlechner,^{16,17}
Andrei Utina,^{16,17} Chris Van Den Broeck,^{16,26} and Teng Zhang^{9,17}

¹Gran Sasso Science Institute (GSSI), I-67100 L'Aquila, Italy

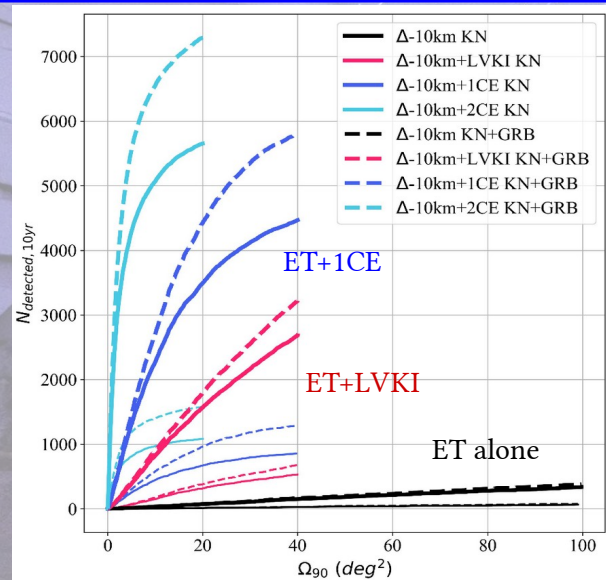
²INFN, Laboratori Nazionali del Gran Sasso, I-67100 Assergi, Italy

*Corresponding author.

Extensive study of networks: Cosmic Explorer and LVKI

NETWORKS	INJECTION SETS			
	α	# events	EOS	Mass distr.
ET-Δ-10km	0.5	71338	APR4	Gaussian
ET- Δ -10km + LVKI				Uniform
ET- Δ -10km + 1CE			BLh	Gaussian
ET- Δ -10km + 2CE				Uniform
ET-2L-15km	1.0	363560	APR4	Gaussian
ET-2L-15km + LVKI				Uniform
ET-2L-15km + 1CE			BLh	Gaussian
ET-2L-15km + 2CE				Uniform

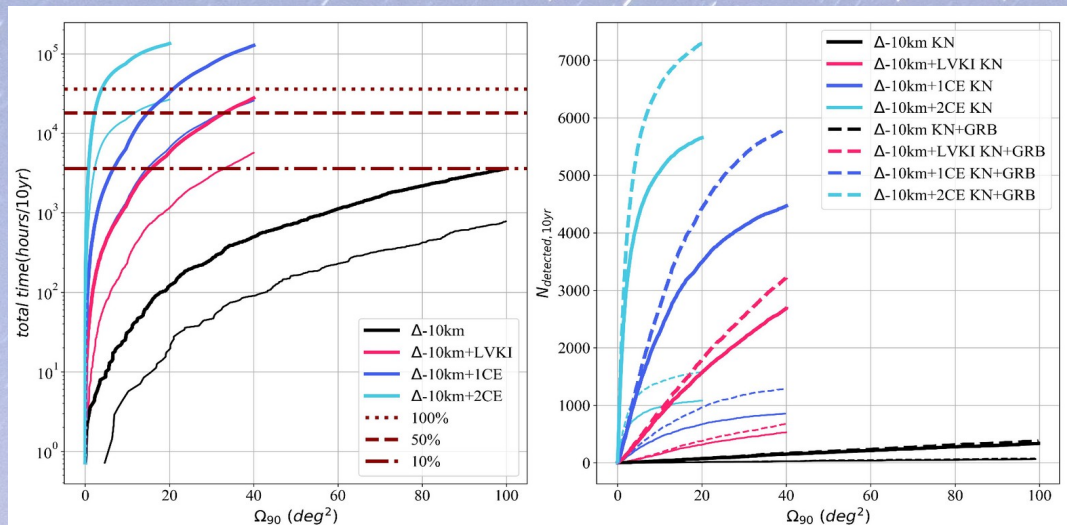
Networks of 3G (with 2G+) are central to harvesting **KNe** and **optical counterparts** in statistical numbers with **Rubin**



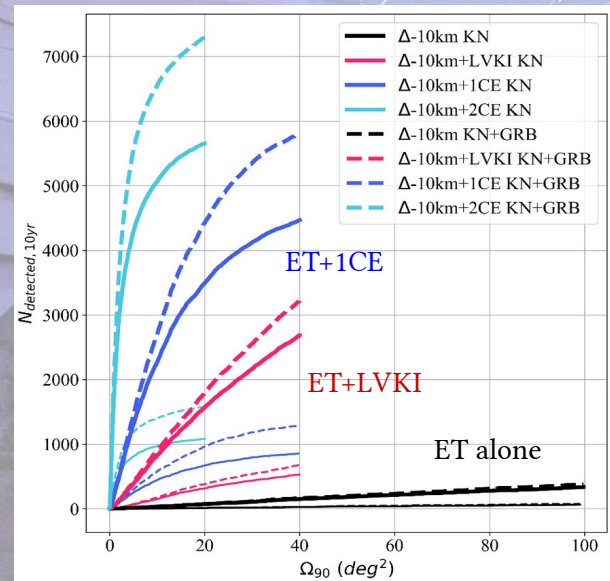
Loffredo, Hazra, Dupletsa, Branchesi et al.



Extensive study of networks: Cosmic Explorer and LVKI



Choose your own adventure



Loffredo, Hazra, Dupletsa, Branchesi et al.

